PES UNIVERSITY

Department of Computer Science & Engineering

Information Security

UE23CS343BB6

Malware Assignment Submission

Name of the Student	Pranav Hemanth
SRN	PES1UG23CS433
Section	G
Department	CSE
Campus	RR

Name of the Student	Sampriti Saha
SRN	PES1UG23CS505
Section	I
Department	CSE
Campus	RR

Malware Assignment

Department of Computer Science & Engineering
Information Security

UE23CS343BB6

Part A: In class malware analysis of satan.exe

Files as part of this lab setup:

This lab uses a malware zip:

```
TEXT TOO SMALL? Run set-scaling auto or set-scaling 2x to adjust.

remnux@remnux:~$ cd Downloads/
remnux@remnux:~/Downloads$ ls
malware.7z
remnux@remnux:~/Downloads$ ls
README.txt  malware.7z
remnux@remnux:~/Downloads$
```

Lab Setup

Step1: Download malware from the link below:

1) https://drive.google.com/drive/folders/1LDCuWu-6gFGtZiIYluGd8BHDaURegvOo
2) Install REMnux

Part 1: Static Analysis

Step2: Statically analyze looking at the bytes of the file:

3) Command: xxd Readme.txt | head

```
remnux@remnux:~/Downloads$ xxd README.txt  | head
00000000: 4749 4638 3961 8002 2602 f780 0000 0000  GIF89a..&.......
00000010: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000020: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000030: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000040: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000050: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000060: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000070: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000080: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000090: 0000 0000 0000 0000 0000 0000 0000 0000  ................
remnux@remnux:~/Downloads$
```

Explanation:
We observe that the file is a gif

Step3: inspect the kind of file:

4) Command: file Readme.txt

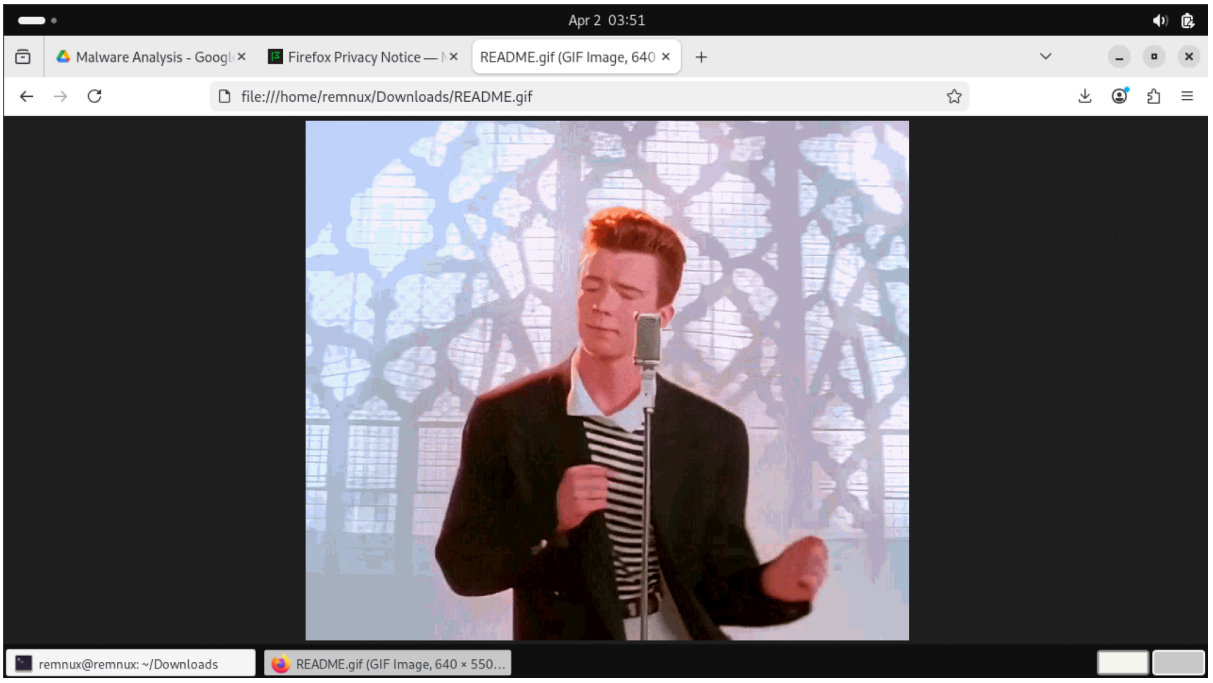
```
remnux@remnux:~/Downloads$ file README.txt
README.txt: GIF image data, version 89a, 640 x 550
remnux@remnux:~/Downloads$
```

Step4: Inspect the file in its original format:

5) Command: mv Readme.txt Readme.gif

```
remnux@remnux:~/Downloads$ mv README.txt README.gif
remnux@remnux:~/Downloads$
```

6) Command: open the file



Step5: Unzip the file:

7) Command: Run 7z x malware.7z
8) Password: infected

```
remnux@remnux:~/Downloads$ 7z x malware.7z
]
7-Zip 23.01 (x64) : Copyright (c) 1999-2023 Igor Pavlov : 2023-06-20
 64-bit locale=C.UTF-8 Threads:2 OPEN_MAX:1024

Scanning the drive for archives:
1 file, 140138 bytes (137 KiB)

Extracting archive: malware.7z
--
Path = malware.7z
Type = 7z
Physical Size = 140138
Headers Size = 154
Method = LZMA2:192k BCJ 7zAES
Solid = -
Blocks = 1


Enter password (will not be echoed):
Everything is Ok

Size:       189331
Compressed: 140138
remnux@remnux:~/Downloads$ ls
README.gif  malware.7z  satan.exe
remnux@remnux:~/Downloads$
```

Step6: Inspect satan.exe:

9) Command: file satan.exe

```
remnux@remnux:~/Downloads$ file satan.exe
satan.exe: PE32 executable (GUI) Intel 80386, for MS Windows, 4 sections
remnux@remnux:~/Downloads$
```

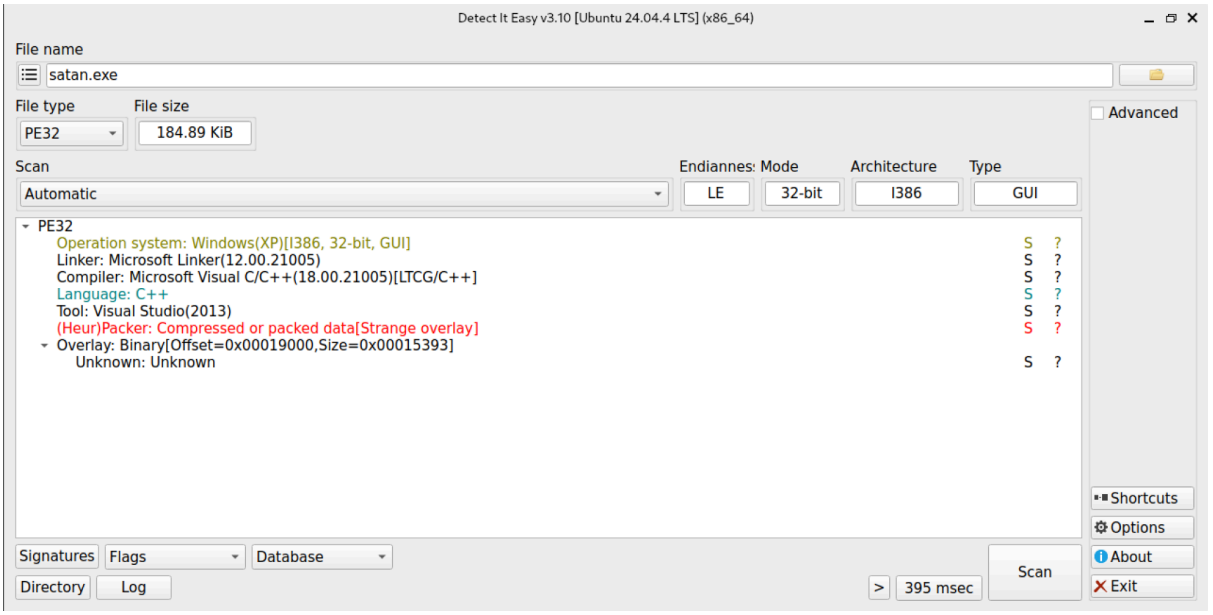
Step7: find the hash of the malware - hashes are used to identify malwares:

10) Command: sha256sum satan.exe

```
remnux@remnux:~/Downloads$ sha256sum satan.exe
28f299818268701eb239974b74b3d80fc28e2d14cc0f45a52a9bc62be784d170  satan.exe
remnux@remnux:~/Downloads$
```

Step8: deep inspect the file with die (detect it easy):

11) Command: die satan.exe



Step9: linux utility for file information:

12)Command: peframe satan.exe

```
remnux@remnux:~/Downloads$ peframe satan.exe

------------------------------- File Information (time: 0:00:02.328401) -------------------------------
filename          satan.exe
filetype          PE32 executable (GUI) Intel 80386, for MS Windows, 4 sections
filesize          189331
md5               f592e7faba96a23ee25ff25f3779f44f
sha1              66656b81b43130ddd6774845c249700ffb06cead
sha256            28f299818268701eb239974b74b3d80fc28e2d14cc0f45a52a9bc62be784d170
virustotal        X-Apikey header is missing
imagebase         0x400000
entrypoint        0x13b9
imphash           65e9607e6f28a7852bb41a6e2e439a92
datetime          2017-02-23 19:28:24
dll               False
directories       import, tls, relocations
sections          .data, .reloc, .text *, .rdata *
features          antidbg, crypto

---------------------------------------------- Yara Plugins ----------------------------------------------
IsPE32
IsWindowsGUI
IsPacked
HasOverlay
HasRichSignature
Advapi Hash API
CRC32 poly Constant

---------------------------------------------- Behavior ----------------------------------------------
SEH  vectored
```

Step10: Check for strings. Strings won't work as malware code is generally in different encodings. We use floss instead:

13)Command: floss satan.exe

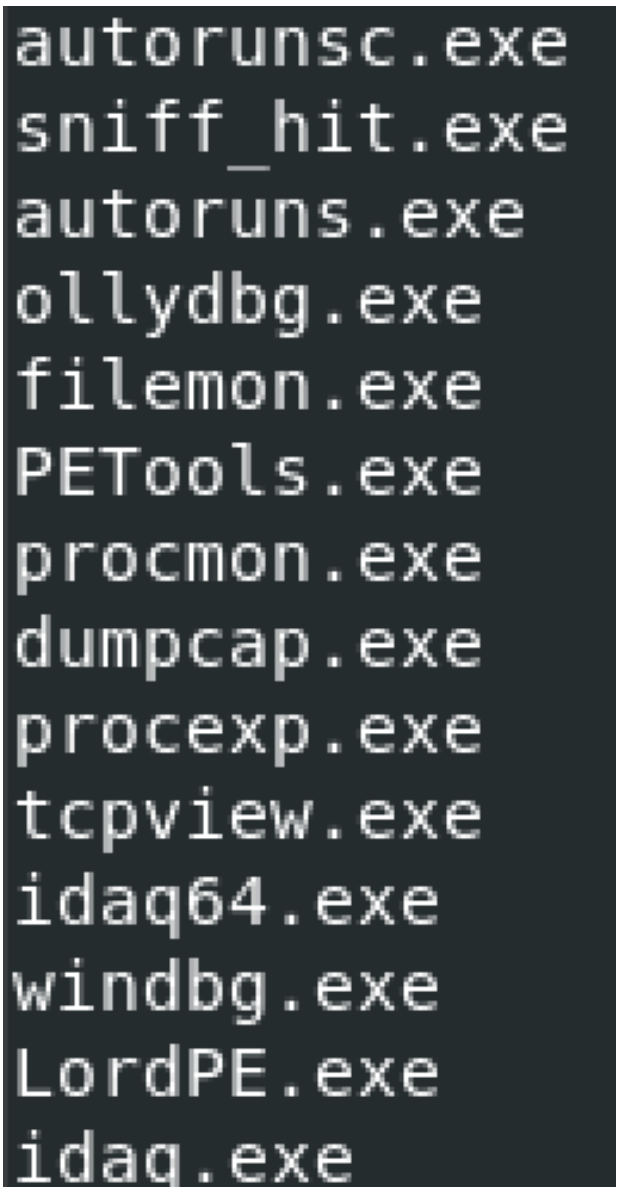
```
+-------------------------------------+
| FLOSS STATIC STRINGS: UTF-16LE (3) |
+-------------------------------------+

Microsoft Enhanced Cryptographic Provider v1.0
Microsoft Enhanced RSA and AES Cryptographic Provider (Prototype)
Microsoft Enhanced RSA and AES Cryptographic Provider


 ___________________________
  FLOSS STACK STRINGS (0)
 ___________________________




 ___________________________
  FLOSS TIGHT STRINGS (3)
 ___________________________

w4ED
mjsB
MQ1a
```

Observation: Malware is heavily encrypted. Could indicate a ransomware

```
FLOSS DECODED STRINGS (107)

netmon.exe
WinDbgFrameClass
\SAMPLE
filemon.exe
VIRUS
SbieDll.dll
ObsidianGUI
PETools.exe
ntdll.dll
SANDBOX
procmon.exe
dbghelp.dll
LordPE.exe
avghookx.dll
C:\InsideTm
snxhk.dll
procexp.exe
SysInspector.exe
Immunity Debugger
MALWARE
avghooka.dll
api_log.dll
idaq.exe
proc_analyser.exe
MALTEST
dir_watch.dll
idaq64.exe
TEQUILABOOMBOOM
sysAnalyzer.exe
```

```
vmcheck.dll
ollydbg.exe
devenv.exe
ImmunityDebugger.exe
kernel32.dll
Zeta Debugger
sample.ex
wpespy.dll
sniff_hit.exe
ProcessHacker.exe
Wireshark.exe
pstorec.dll
windbg.exe
tcpview.exe
OLLYDBG
dwmapi.dll
user32.dll
ntdll.d
csrss.e
dumpcap.exe
joeboxcontrol.exe
autoruns.exe
Rock Debugger
HookExplorer.exe
joeboxserver.exe
autorunsc.exe
ImportREC.exe
NtGetContextThread
LdrLoadDll
NtSetContextThread
NtUnmapViewOfSection
NtResumeThread
```

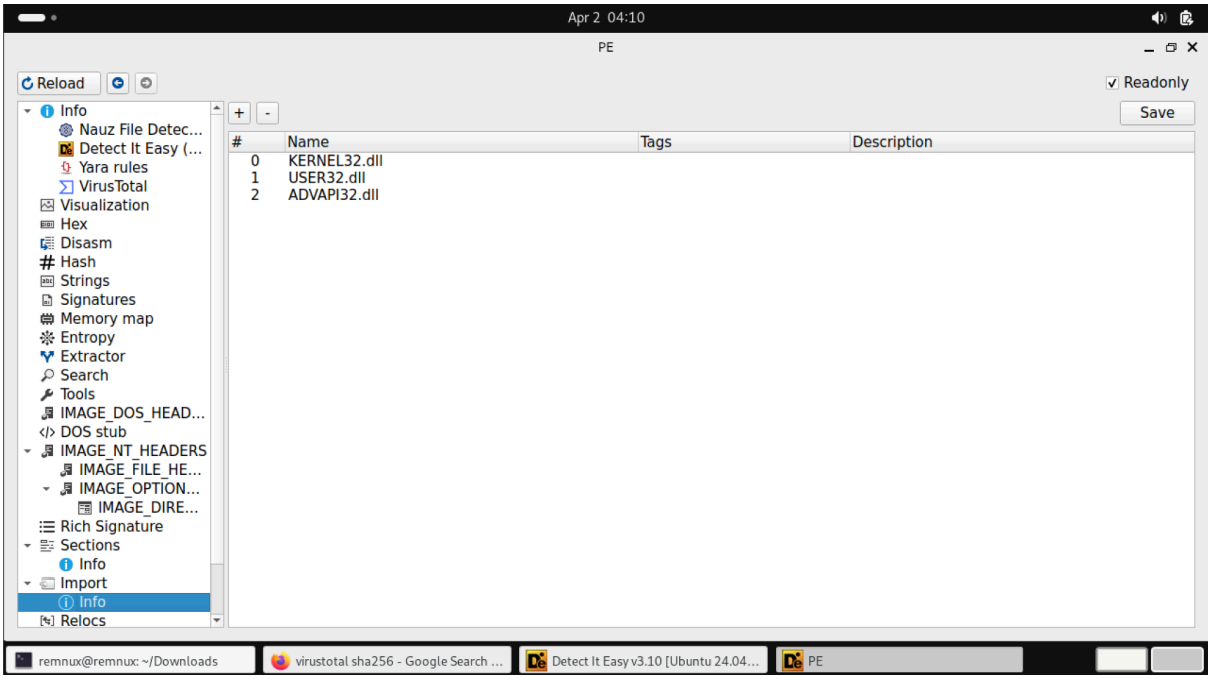
```
NtClose
ntdll.dll
NtQueryInformationProcess
kernel32.dll
VirtualAllocEx
CreateProces
IsDebuggerPresent
CheckRemoteDebuggerPresent
poAe
NtTerminateProcess
BlockInput
NtReadVirtualMemory
PuAt
wine_get_unix_file_name
NtWriteVirtualMemory
dir_watch.dll
vmcheck.dll
dbghelp.dll
SbieDll.dll
pstorec.dll
api_log.dll
wpespy.dll
snxhk.dll
ImmunityDebugger.exe
joeboxcontrol.exe
ProcessHacker.exe
proc_analyser.exe
HookExplorer.exe
SysInspector.exe
sysAnalyzer.exe
ImportREC.exe
Wireshark.exe
```

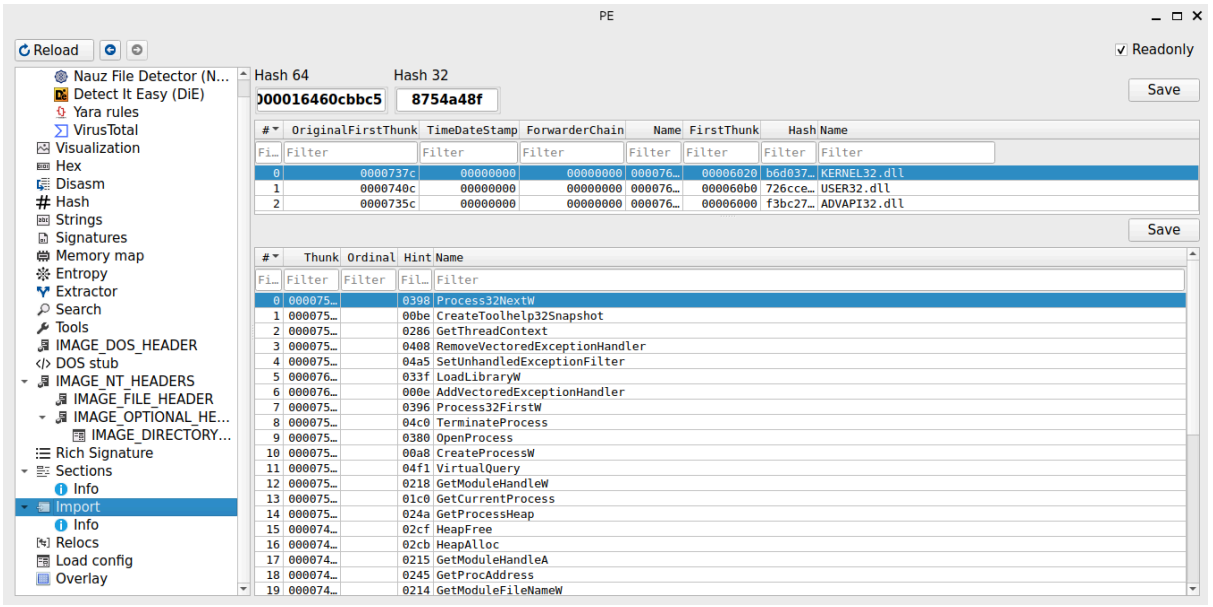


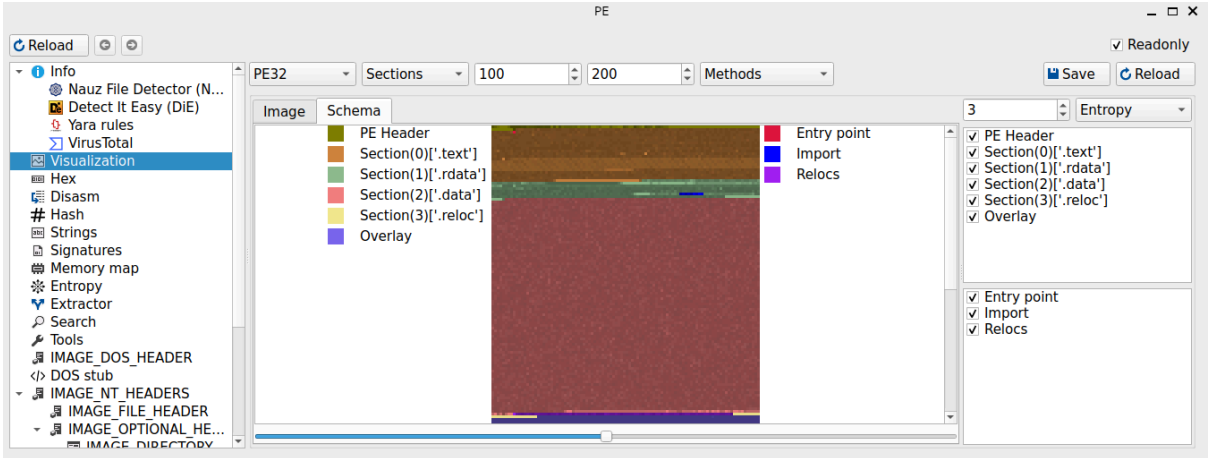
Step11: Find imports using die:

14)Command: die satan.exe

Jan - May 2026 Assignment SUBMISSION_UE23CS343BB6







Part 2: Dynamic Analysis

(IMPORTANT): Create a snapshot of vm as after dynamic analysis, the vm will be corrupted:

Step12: Move satan.exe to upper dir

15)Command: mv satan.exe ../

```
remnux@remnux:~/Downloads$ mv satan.exe  ../
remnux@remnux:~/Downloads$ ls
README.gif  malware.7z
remnux@remnux:~/Downloads$ cd ..
remnux@remnux:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  satan.exe  snap
remnux@remnux:~$
```

Step13: Setup wine:

16)Command: winecfg

```
remnux@remnux:~$ winecfg
002c:fixme:winediag:loader_init wine-staging 11.5 is a testing version containing experimental patches.
002c:fixme:winediag:loader_init Please mention your exact version when filing bug reports on winehq.org.
libEGL warning: DRI3 error: Could not get DRI3 device
libEGL warning: Ensure your X server supports DRI3 to get accelerated rendering
VMware: No 3D enabled (0, Success).
libEGL warning: egl: failed to create dri2 screen
VMware: No 3D enabled (0, Success).
libEGL warning: egl: failed to create dri2 screen
VMware: No 3D enabled (0, Success).
libEGL warning: egl: failed to create dri2 screen
00b0:err:ntoskrnl:ZwLoadDriver failed to create driver L"\\Registry\\Machine\\System\\CurrentControlSet\\Services\\winebth": c00000e5
003c:fixme:service:scmdatabase_autostart_services Auto-start service L"winebth" failed to start: 1359
0024:fixme:winediag:loader_init wine-staging 11.5 is a testing version containing experimental patches.
0024:fixme:winediag:loader_init Please mention your exact version when filing bug reports on winehq.org.
remnux@remnux:~$ 012c:fixme:winediag:loader_init wine-staging 11.5 is a testing version containing experimental patches.
012c:fixme:winediag:loader_init Please mention your exact version when filing bug reports on winehq.org.

remnux@remnux:~$
```

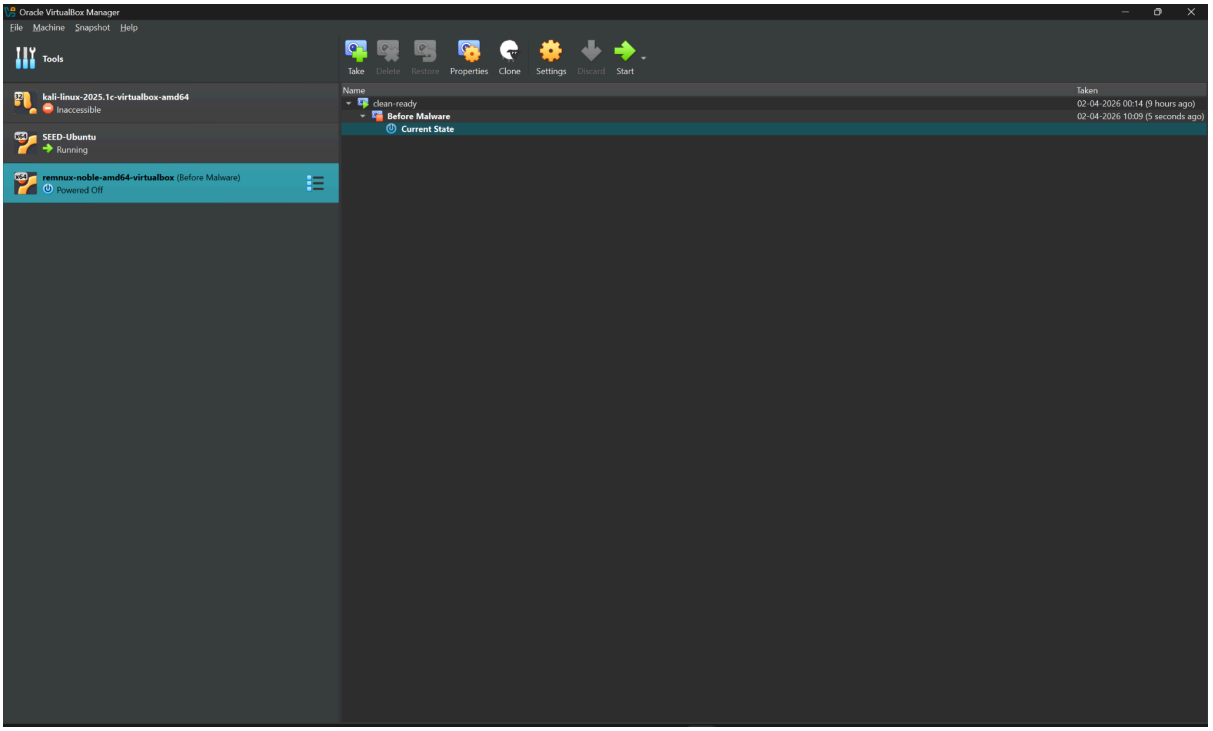
Step14: Verify setup:

```
remnux@remnux:~$ cd .wine/
remnux@remnux:~/.wine$ ls
dosdevices  drive_c  system.reg  user.reg  userdef.reg
remnux@remnux:~/.wine$ cd drive_c/
remnux@remnux:~/.wine/drive_c$ ls
'Program Files'  'Program Files (x86)'  ProgramData  users  windows
remnux@remnux:~/.wine/drive_c$ cd  users/
remnux@remnux:~/.wine/drive_c/users$ ls
Public  remnux
remnux@remnux:~/.wine/drive_c/users$ cd remnux/
remnux@remnux:~/.wine/drive_c/users/remnux$ ls
 AppData   Contacts   Desktop   Documents   Downloads   Favorites   Links   Music   Pictures  'Saved Games'   Searches   Videos
remnux@remnux:~/.wine/drive_c/users/remnux$
```

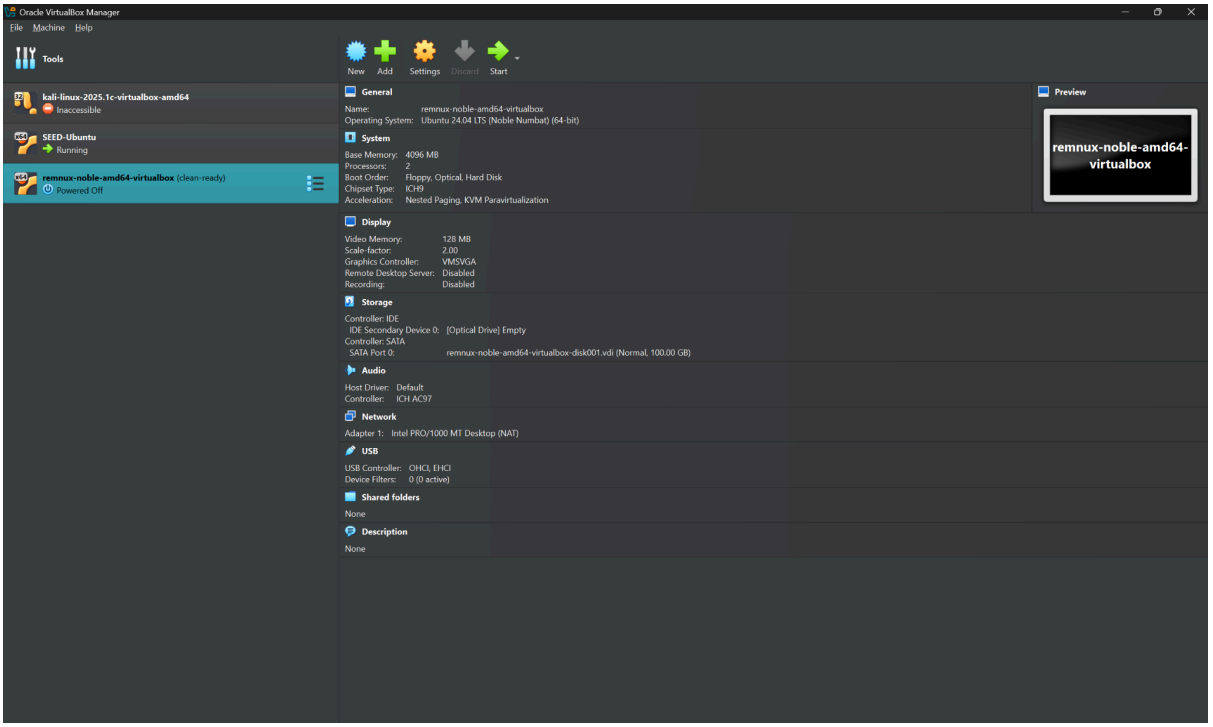
Step15: Power the vm off:

17)Command: poweroff

Step16: Take the snapshot:



Step17: See the snapshot details:



Step18: Install inotifywait:

18)Command: sudo apt install inotifywait-tools

```
remnux@remnux:~$ sudo apt install inotify-tools
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
inotify-tools is already the newest version (3.22.6.0-4).
0 upgraded, 0 newly installed, 0 to remove and 29 not upgraded.
```

Step19: Setup a directory watch:

19)Command: inotifywait -m -r ~/.wine --format '%T %e %w%f' --timefmt '%H:%M:%S' -e create,modify,delete,move

```
remnux@remnux:~$ inotifywait -m -r ~/.wine --format '%T %e %w%f' --timefmt '%H:%M:%S' -e create,modi
fy,delete,move
Setting up watches.  Beware: since -r was given, this may take a while!
Watches established.
```

Step20: run tcpdump:

20)Command: sudo tcpdump -i any -n -w satan_capture.pcap 'not port 22'

```
remnux@remnux:~$ sudo tcpdump -i any -n -w satan_capture.pcap 'not port 22'
tcpdump: data link type LINUX_SLL2
tcpdump: listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
```

Step21: Run wine:

21)Command: wine satan.exe

```
remnux@remnux:~$ wine satan.exe
002c:fixme:winediag:loader_init wine-staging 11.5 is a testing version containing experimental patch
es.
002c:fixme:winediag:loader_init Please mention your exact version when filing bug reports on winehq.
org.
libEGL warning: DRI3 error: Could not get DRI3 device
libEGL warning: Ensure your X server supports DRI3 to get accelerated rendering
VMware: No 3D enabled (0, Success).
libEGL warning: egl: failed to create dri2 screen
VMware: No 3D enabled (0, Success).
libEGL warning: egl: failed to create dri2 screen
VMware: No 3D enabled (0, Success).
libEGL warning: egl: failed to create dri2 screen
00b0:err:ntoskrnl:ZwLoadDriver failed to create driver L"\\Registry\\Machine\\System\\CurrentControl
Set\\Services\\winebth": c00000e5
003c:fixme:service:scmdatabase_autostart_services Auto-start service L"winebth" failed to start: 135
9
0024:fixme:winediag:loader_init wine-staging 11.5 is a testing version containing experimental patch
es.
0024:fixme:winediag:loader_init Please mention your exact version when filing bug reports on winehq.
org.
0024:fixme:keyboard:BlockInput (1): stub
remnux@remnux:~$ 0120:fixme:winediag:loader_init wine-staging 11.5 is a testing version containing e
xperimental patches.
0120:fixme:winediag:loader_init Please mention your exact version when filing bug reports on winehq.
org.
0120:fixme:volume:GetVolumeNameForVolumeMountPointW Mounted Folders are not yet supported
0130:fixme:winediag:loader_init wine-staging 11.5 is a testing version containing experimental patch
es.
0130:fixme:winediag:loader_init Please mention your exact version when filing bug reports on winehq.
```

inotifywait watch:

```
remnux@remnux:~$ inotifywait -m -r ~/.wine --format '%T %e %w%f' --timefmt '%H:%M:%S' -e create,modi
fy,delete,move
Setting up watches.  Beware: since -r was given, this may take a while!
Watches established.
04:53:01 CREATE,ISDIR /home/remnux/.wine/drive_c/users/remnux/AppData/Roaming/Soom
04:53:01 CREATE /home/remnux/.wine/drive_c/users/remnux/AppData/Roaming/Soom/efin.exe
04:53:01 MODIFY /home/remnux/.wine/drive_c/users/remnux/AppData/Roaming/Soom/efin.exe
04:53:01 MODIFY /home/remnux/.wine/drive_c/users/remnux/AppData/Roaming/Soom/efin.exe
04:53:01 MODIFY /home/remnux/.wine/drive_c/users/remnux/AppData/Roaming/Soom/efin.exe
04:53:01 CREATE /home/remnux/.wine/drive_c/users/remnux/AppData/Local/Temp/tmp_6496b586.bat
04:53:01 MODIFY /home/remnux/.wine/drive_c/users/remnux/AppData/Local/Temp/tmp_6496b586.bat
04:53:01 MODIFY /home/remnux/.wine/drive_c/users/remnux/AppData/Local/Temp/tmp_6496b586.bat
04:53:02 DELETE /home/remnux/.wine/drive_c/users/remnux/AppData/Local/Temp/tmp_6496b586.bat
04:53:14 CREATE /home/remnux/.wine/drive_c/users/remnux/AppData/Local/Microsoft/Windows/INetCache/Co
ntent.IE5/YWCS32SZ/U3P1XWNO
04:53:14 MODIFY /home/remnux/.wine/drive_c/users/remnux/AppData/Local/Microsoft/Windows/INetCache/Co
ntent.IE5/YWCS32SZ/U3P1XWNO
04:53:14 CREATE /home/remnux/.wine/drive_c/users/remnux/AppData/Local/Microsoft/Windows/INetCache/Co
ntent.IE5/6B4IA5J6/L62BXWNO
04:53:14 MODIFY /home/remnux/.wine/drive_c/users/remnux/AppData/Local/Microsoft/Windows/INetCache/Co
ntent.IE5/6B4IA5J6/L62BXWNO
04:53:14 MODIFY /home/remnux/.wine/drive_c/users/remnux/AppData/Local/Microsoft/Windows/INetCache/Co
ntent.IE5/6B4IA5J6/L62BXWNO
04:53:14 DELETE /home/remnux/.wine/drive_c/users/remnux/AppData/Local/Microsoft/Windows/INetCache/Co
ntent.IE5/YWCS32SZ/U3P1XWNO
04:53:29 CREATE /home/remnux/.wine/regbac0000.tmp
```

tcpdump:

```
remnux@remnux:~$ sudo tcpdump -i any -n -w satan_capture.pcap 'not port 22'
tcpdump: data link type LINUX_SLL2
tcpdump: listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
^C168 packets captured
174 packets received by filter
0 packets dropped by kernel
remnux@remnux:~$ tcpdump -r satan_capture.pcap
reading from file satan_capture.pcap, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144
Warning: interface names might be incorrect
04:53:12.241710 lo    In  IP localhost.46891 > _localdnsstub.domain: 61677+ [1au] A? xhj4hypdsb3jozw
n.onion.pw. (54)
04:53:12.241950 enp0s3 Out IP remnux.41793 > 10.0.2.3.domain: 46349+ [1au] A? xhj4hypdsb3jozwn.onion
.pw. (54)
04:53:12.296002 enp0s3 In  IP 10.0.2.3.domain > remnux.41793: 46349 1/0/1 A 149.28.244.165 (70)
04:53:12.296457 lo    In  IP _localdnsstub.domain > localhost.46891: 61677 1/0/1 A 149.28.244.165 (7
0)
04:53:12.365716 enp0s3 Out ARP, Request who-has _gateway tell remnux, length 28
04:53:12.366126 enp0s3 In  ARP, Reply _gateway is-at 52:55:0a:00:02:02 (oui Unknown), length 50
04:53:12.366132 enp0s3 Out IP remnux.41592 > 149.28.244.165.vultrusercontent.com.http: Flags [S], se
q 144603937, win 64240, options [mss 1460,sackOK,TS val 2325950560 ecr 0,nop,wscale 7], length 0
04:53:12.412006 enp0s3 In  IP 149.28.244.165.vultrusercontent.com.http > remnux.41592: Flags [S.], s
eq 30720001, ack 144603938, win 65535, options [mss 1460], length 0
04:53:12.412074 enp0s3 Out IP remnux.41592 > 149.28.244.165.vultrusercontent.com.http: Flags [.], ac
k 1, win 64240, length 0
04:53:12.413660 enp0s3 Out IP remnux.41592 > 149.28.244.165.vultrusercontent.com.http: Flags [P.], s
eq 1:434, ack 1, win 64240, length 433: HTTP: POST /g.php HTTP/1.1
04:53:12.414093 enp0s3 In  IP 149.28.244.165.vultrusercontent.com.http > remnux.41592: Flags [.], ac
k 434, win 65535, length 0
04:53:13.099739 enp0s3 In  IP 149.28.244.165.vultrusercontent.com.http > remnux.41592: Flags [P.], s
eq 1:384, ack 434, win 65535, length 383: HTTP: HTTP/1.1 301 Moved Permanently
```

Step22: Check the capture for onion:

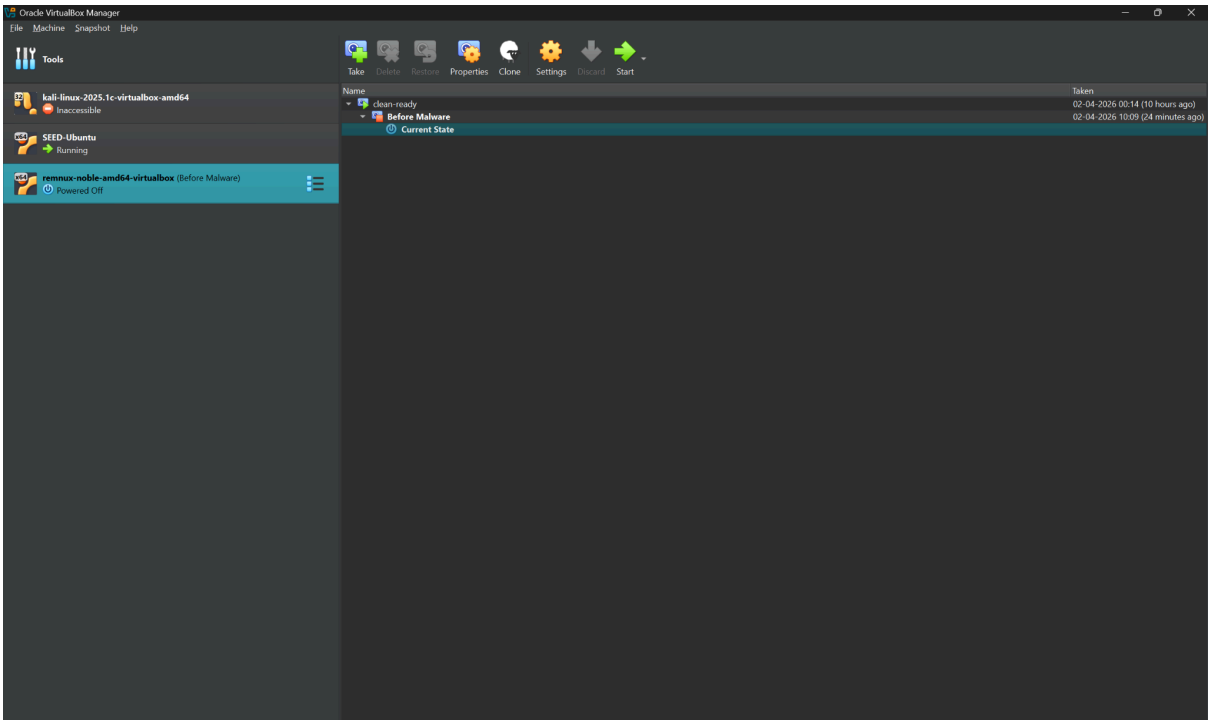
22)Command: tcpdump -R satan_capture.pcap | GREP -iE onion

```
remnux@remnux:~$ tcpdump -r satan_capture.pcap | grep -iE onion
reading from file satan_capture.pcap, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144
Warning: interface names might be incorrect
04:53:12.241710 lo    In  IP localhost.46891 > _localdnsstub.domain: 61677+ [1au] A? xhj4hypdsb3jozwn.onion.pw. (54)
04:53:12.241950 enp0s3 Out IP remnux.41793 > 10.0.2.3.domain: 46349+ [1au] A? xhj4hypdsb3jozwn.onion.pw. (54)
04:53:13.108022 lo    In  IP localhost.38360 > _localdnsstub.domain: 16726+ [1au] A? xhj4hypdsb3jozwn.onion.pw. (54)
remnux@remnux:~$
```

Step23: Inspect the onion domain:

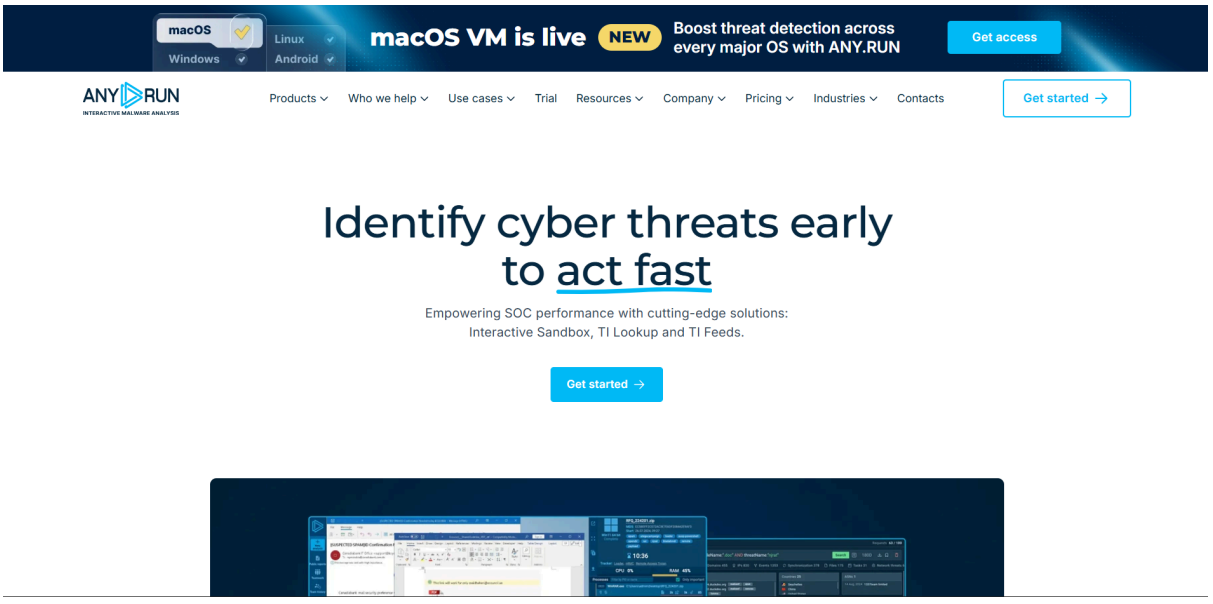
23) xhj4hypdsb3jozwn.onion.pw (Blocked by PES Network)

Step24: Restore snapshot:



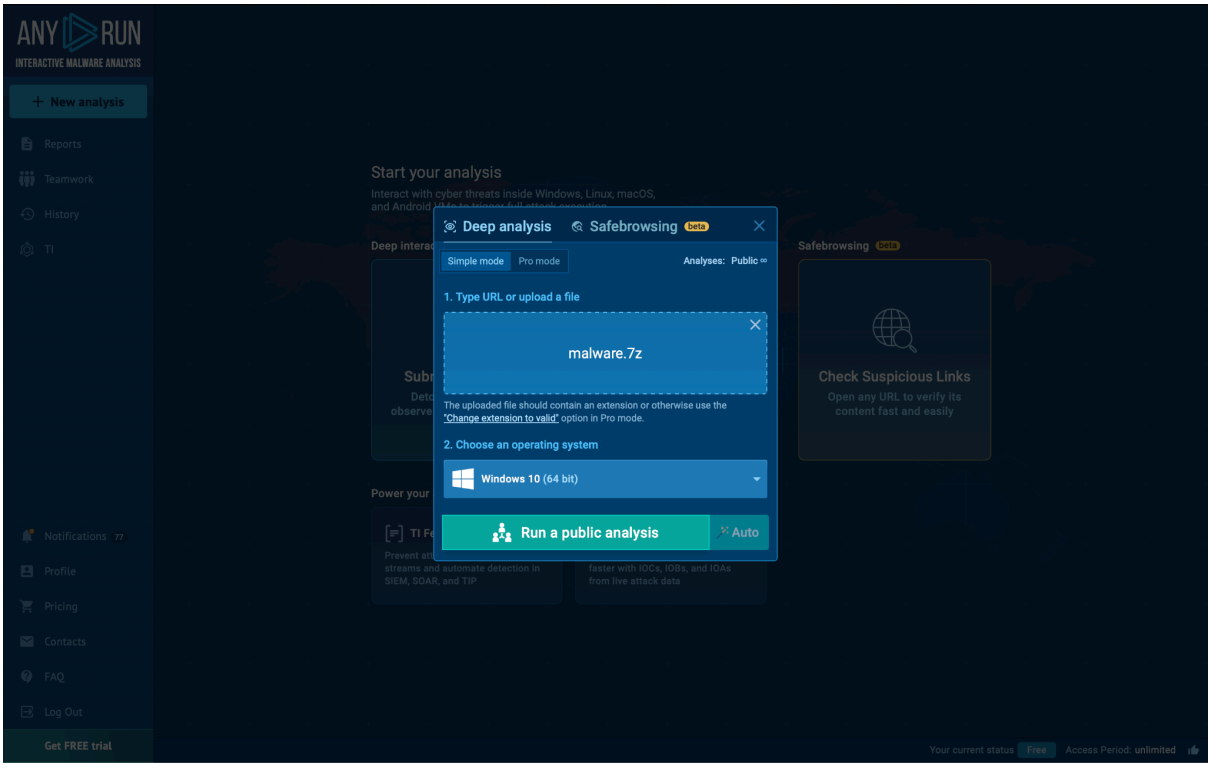
Note: Do it safely online without worrying about the malwares

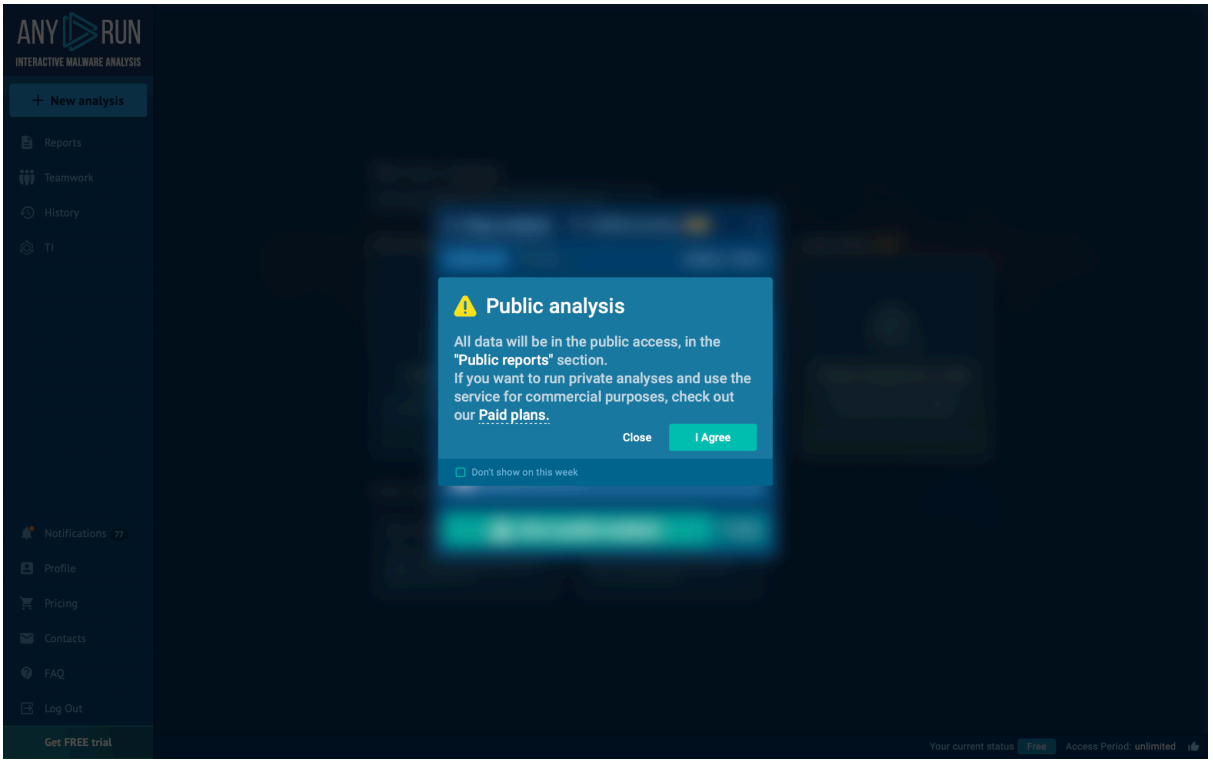
Step25: Do it online on anyrun:

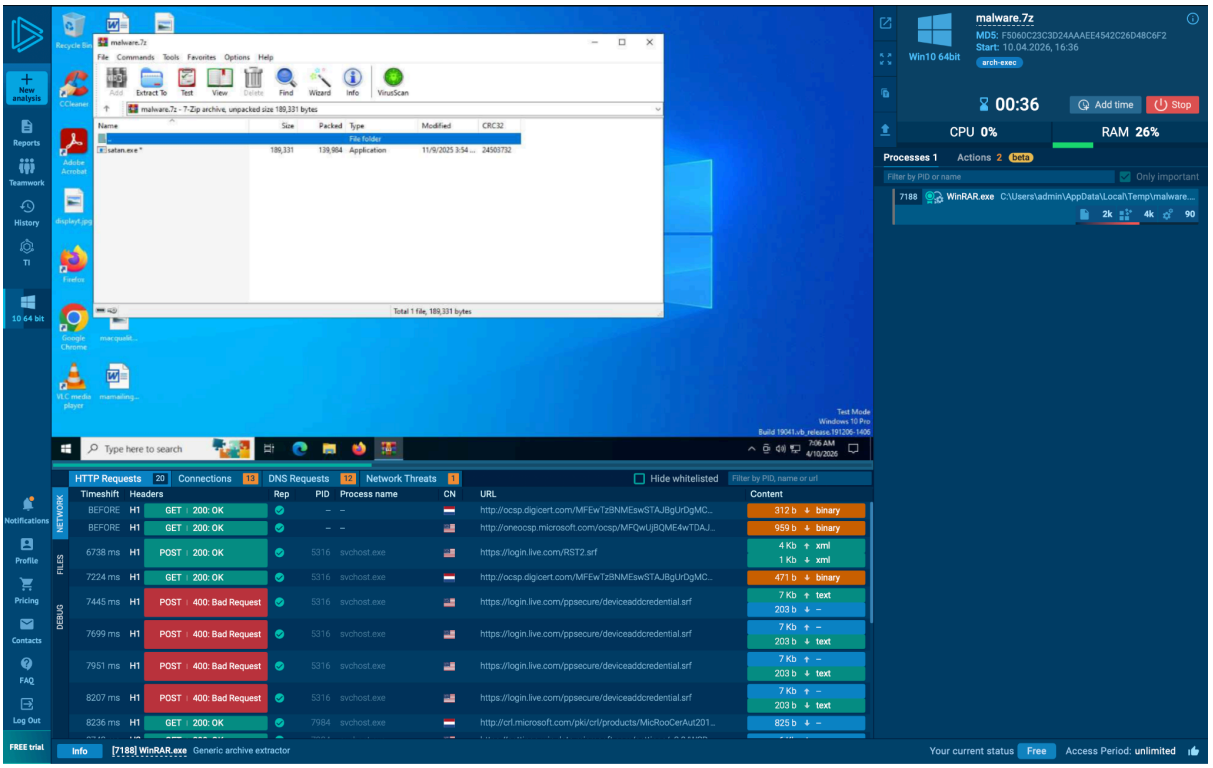


Step26: Open the console:

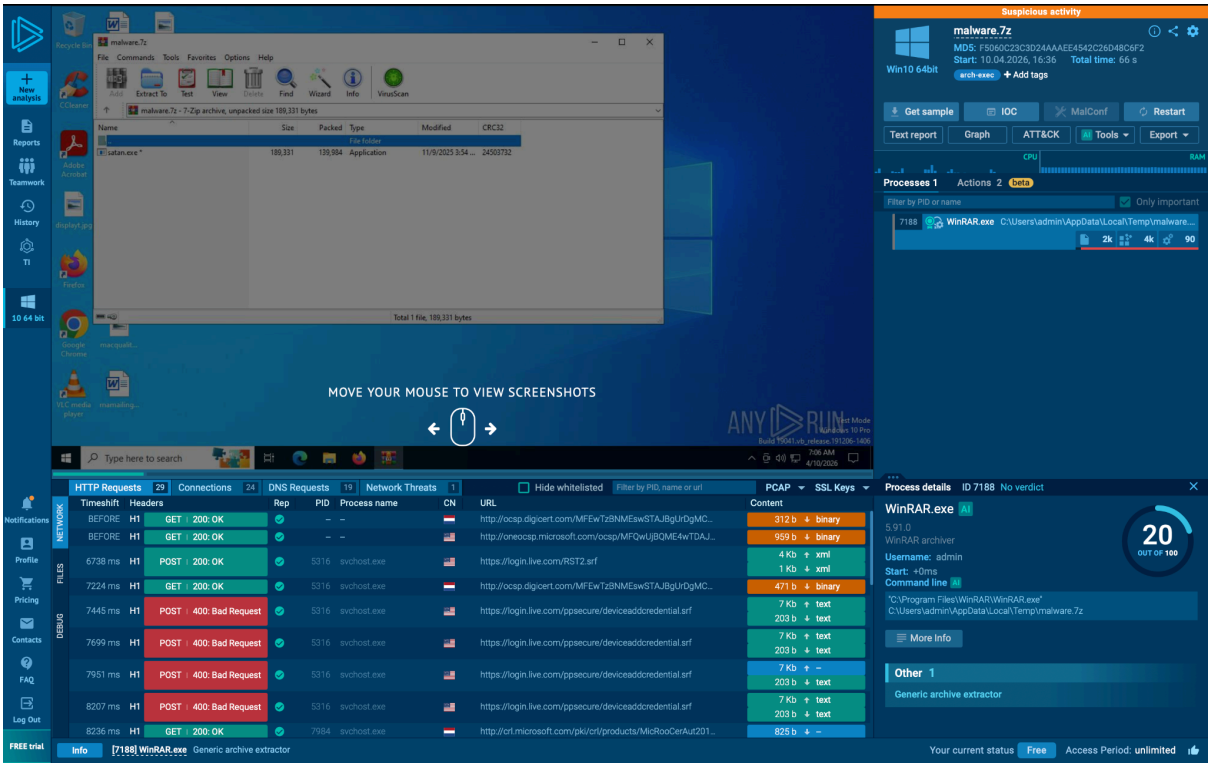
Step26: Upload the malware file to run analysis:



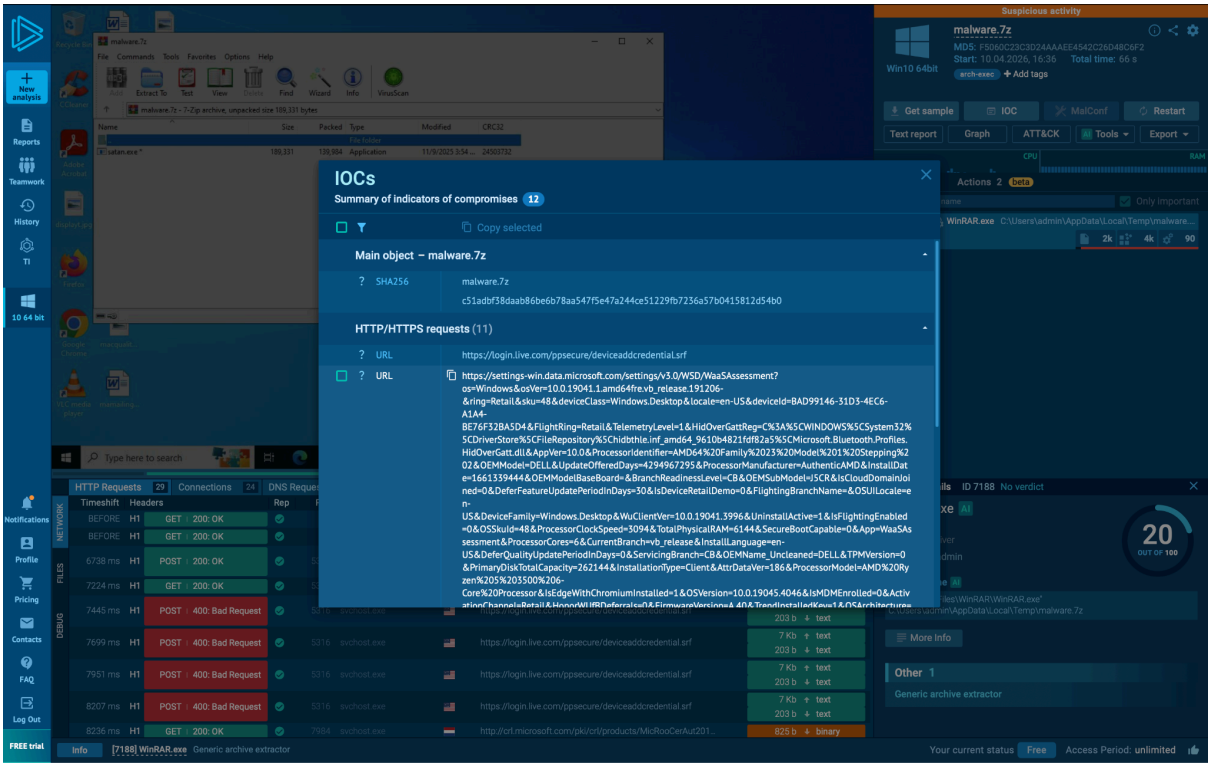




Step27: Observe the processes:



Step28: Observe the IOCs:



Step29: Generate the analysis report:

ANY RUN
INTERACTIVE MALWARE ANALYSIS

General Info

File name:	malware.7z
Full analysis:	https://app.any.run/tasks/241834ac-d0c0-488e-9cfc-befb2bacaf35
Verdict:	Suspicious activity
Analysis date:	April 10, 2026 at 16:36:06
OS:	Windows 10 Professional (build: 19044, 64 bit)
Tags:	arch-exec
MIME:	application/x-7z-compressed
File info:	7-zip archive data, version 0.4
MD5:	F5060C23C3D24AAAEE4542C26D48C6F2
SHA1:	A44C603C33EE39CC8DAC337B720D92D1A19AB17B
SHA256:	C51ADBF38DAAB86BE6B78AA547F5E47A244CE51229FB7236A57B0415812D54B0
SSDEEP:	6144:idZ154zRVFwf9f9SAXV8nsWxO5NWU41wTQ6:idZD4jKfh958nsWxO7g1wTB

Software environment set and analysis options

Launch configuration

Task duration:	60 seconds	Heavy Evasion option:	off	Network geolocation:	off
Additional time used:	none	MITM proxy:	off	Privacy:	Public submission
Fakenet option:	off	Route via Tor:	off	Autoconfirmation of UAC:	on
Network:	on				

Software preset

- Internet Explorer 11.3636.19041.0
- Adobe Acrobat (64-bit) (23.001.20093)
- Adobe Flash Player 32 NPAPI (32.0.0.465)
- Adobe Flash Player 32 PPAPI (32.0.0.465)
- CCleaner (6.20)
- Google Chrome (133.0.6943.127)
- Google Update Helper (1.3.36.51)
- Java 8 Update 271 (64-bit) (8.0.2710.9)
- Java Auto Updater (2.8.271.9)
- Java(TM) SE Development Kit 25.0.2 (64-bit) (25.0.2.0)
- Microsoft Edge (133.0.3065.92)
- Microsoft Office Professional 2019 - de-de (16.0.16026.20146)
- Microsoft Office Professional 2019 - en-us (16.0.16026.20146)
- Microsoft Office Professional 2019 - es-es (16.0.16026.20146)
- Microsoft Office Professional 2019 - it-it (16.0.16026.20146)
- Microsoft Office Professional 2019 - ja-jp (16.0.16026.20146)
- Microsoft Office Professional 2019 - ko-kr (16.0.16026.20146)
- Microsoft Office Professional 2019 - pt-br (16.0.16026.20146)
- Microsoft Office Professional 2019 - tr-tr (16.0.16026.20146)
- Microsoft Office Professionnel 2019 - fr-fr (16.0.16026.20146)
- Microsoft Office профессиональный 2019 - ru-ru (16.0.16026.20146)
- Microsoft OneNote - en-us (16.0.16026.20146)
- Microsoft Update Health Tools (3.74.0.0)
- Microsoft Visual C++ 2013 Redistributable (x64) - 12.0.30501 (12.0.30501.0)
- Microsoft Visual C++ 2013 x64 Additional Runtime - 12.0.21005 (12.0.21005)
- Microsoft Visual C++ 2013 x64 Minimum Runtime - 12.0.21005 (12.0.21005)
- Microsoft Visual C++ 2015-2022 Redistributable (x64) - 14.36.32532 (14.36.32532.0)
- Microsoft Visual C++ 2015-2022 Redistributable (x86) - 14.36.32532 (14.36.32532.0)
- Microsoft Visual C++ 2022 X64 Additional Runtime - 14.36.32532 (14.36.32532)
- Microsoft Visual C++ 2022 X64 Minimum Runtime - 14.36.32532 (14.36.32532)
- Microsoft Visual C++ 2022 X86 Additional Runtime - 14.36.32532 (14.36.32532)
- Microsoft Visual C++ 2022 X86 Minimum Runtime - 14.36.32532 (14.36.32532)
- Mozilla Firefox (x64 en-US) (136.0)
- Mozilla Maintenance Service (136.0)
- Notepad++ (64-bit x64) (7.9.1)
- Office 16 Click-to-Run Extensibility Component (16.0.15726.20202)
- Office 16 Click-to-Run Licensing Component (16.0.16026.20146)
- Office 16 Click-to-Run Localization Component (16.0.15726.20202)

Hotfixes

- Client LanguagePack Package
- DotNetRollup
- DotNetRollup 481
- FodMetadata Package
- Foundation Package
- Hello Face Package
- InternetExplorer Optional Package
- KB5003791
- KB5011048
- KB5015684
- KB5033052
- LanguageFeatures Basic en us Package
- LanguageFeatures Handwriting en us Package
- LanguageFeatures OCR en us Package
- LanguageFeatures Speech en us Package
- LanguageFeatures TextToSpeech en us Package
- MSPaint FoD Package
- MediaPlayer Package
- Microsoft OneCore ApplicationModel Sync Desktop FOD Package
- Microsoft OneCore DirectX Database FOD Package
- NetFx3 OnDemand Package
- Notepad FoD Package
- OpenSSH Client Package
- PowerShell ISE FOD Package
- Printing PMCPPC FoD Package
- Printing WFS FoD Package
- ProfessionalEdition
- QuickAssist Package
- RollupFix
- ServicingStack
- ServicingStack 3989
- StepsRecorder Package
- TabletPCMath Package
- UserExperience Desktop Package
- WordPad FoD Package

- Office 16 Click-to-Run Localization Component (16.0.15928.20198)
- PowerShell 7-x64 (7.3.5.0)
- Update for Windows 10 for x64-based Systems (KB4023057) (2.59.0.0)
- Update for Windows 10 for x64-based Systems (KB4023057) (2.63.0.0)
- Update for Windows 10 for x64-based Systems (KB4480730) (2.55.0.0)
- Update for Windows 10 for x64-based Systems (KB5001716) (8.93.0.0)
- VLC media player (3.0.11)
- WinRAR 5.91 (64-bit) (5.91.0)
- Windows PC Health Check (3.6.2204.08001)

Behavior activities

MALICIOUS

No malicious indicators.

SUSPICIOUS

No suspicious indicators.

INFO

Generic archive extractor

- WinRAR.exe (PID: 7188)

Malware configuration

No Malware configuration.

Static information

TRiD

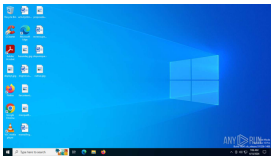
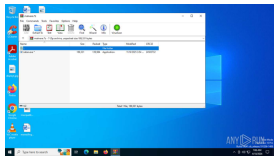
.7z | 7-Zip compressed archive (v0.4) (57.1)
.7z | 7-Zip compressed archive (gen) (42.8)

EXIF

ZIP

FileVersion:	7z v0.04
ModifyDate:	2025:11:09 08:54:50+00:00
ArchivedFileName:	satan.exe

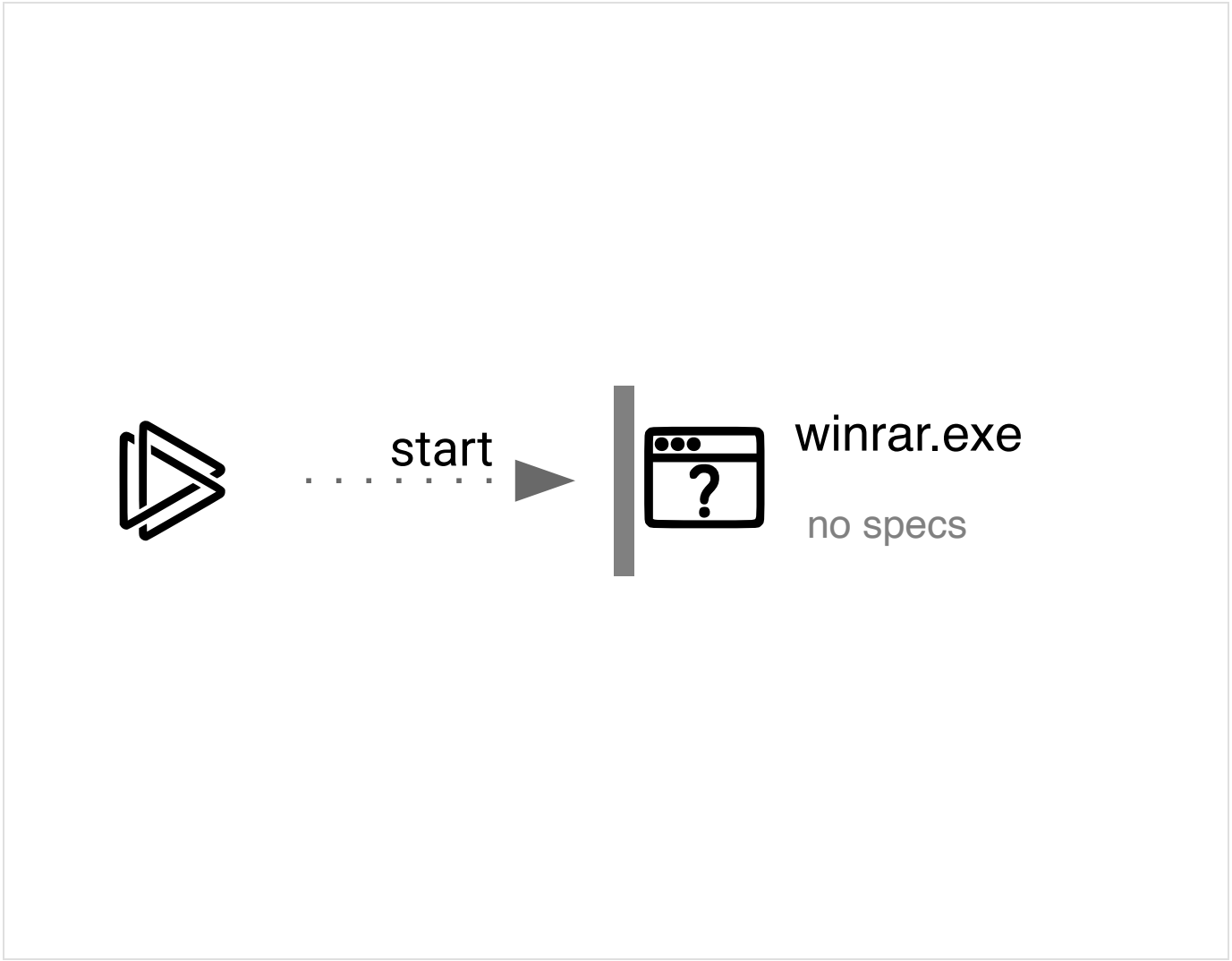
Video and screenshots

Processes

Total processes	Monitored processes	Malicious processes	Suspicious processes
134	1	0	0

Behavior graph



Specs description

Program did not start	Low-level access to the HDD	Process was added to the startup	Debug information is available
Probably Tor was used	Behavior similar to spam	Task has injected processes	Executable file was dropped
Known threat	RAM overrun	Network attacks were detected	Integrity level elevation
Connects to the network	CPU overrun	Process starts the services	System was rebooted
Task contains several apps running	Application downloaded the executable file	Actions similar to stealing personal data	Task has apps ended with an error
File is detected by antivirus software	Inspected object has suspicious PE structure	Behavior similar to exploiting the vulnerability	Task contains an error or was rebooted
The process has the malware config			

Process information

PID	CMD	Path	Indicators	Parent process
7188	"C:\Program Files\WinRAR\WinRAR.exe" C:\Users\admin\AppData\Local\Temp\malware.7z	C:\Program Files\WinRAR\WinRAR.exe	–	explorer.exe

Information			
User:	admin	Company:	Alexander Roshal
Integrity Level:	MEDIUM	Description:	WinRAR archiver
Version:	5.91.0		

Registry activity

Total events	Read events	Write events	Delete events
1 617	1 609	8	0

Modification events

(PID) Process:	(7188) WinRAR.exe	Key:	HKEY_CURRENT_USER\SOFTWARE\WinRAR\ArcHistory
Operation:	write	Name:	3
Value:	C:\Users\admin\Desktop\chromium_ext.zip		
(PID) Process:	(7188) WinRAR.exe	Key:	HKEY_CURRENT_USER\SOFTWARE\WinRAR\ArcHistory
Operation:	write	Name:	2
Value:	C:\Users\admin\Desktop\omni_23_10_2024_.zip		
(PID) Process:	(7188) WinRAR.exe	Key:	HKEY_CURRENT_USER\SOFTWARE\WinRAR\ArcHistory
Operation:	write	Name:	1
Value:	C:\Users\admin\Downloads\chromium_build 1.zip		
(PID) Process:	(7188) WinRAR.exe	Key:	HKEY_CURRENT_USER\SOFTWARE\WinRAR\ArcHistory
Operation:	write	Name:	0
Value:	C:\Users\admin\AppData\Local\Temp\malware.7z		
(PID) Process:	(7188) WinRAR.exe	Key:	HKEY_CURRENT_USER\SOFTWARE\WinRAR\FileList\FileColumnWidths
Operation:	write	Name:	name
Value:	120		
(PID) Process:	(7188) WinRAR.exe	Key:	HKEY_CURRENT_USER\SOFTWARE\WinRAR\FileList\FileColumnWidths
Operation:	write	Name:	size
Value:	80		
(PID) Process:	(7188) WinRAR.exe	Key:	HKEY_CURRENT_USER\SOFTWARE\WinRAR\FileList\FileColumnWidths
Operation:	write	Name:	type
Value:	120		
(PID) Process:	(7188) WinRAR.exe	Key:	HKEY_CURRENT_USER\SOFTWARE\WinRAR\FileList\FileColumnWidths
Operation:	write	Name:	mtime
Value:	100		

Files activity

Executable files	Suspicious files	Text files	Unknown types
0	0	0	0

Dropped files

No data

Network activity

HTTP(S) requests	TCP/UDP connections	DNS requests	Threats
29	24	19	1

HTTP requests

PID	Process	Method	HTTP Code	IP	URL	CN	Type	Size	Reputation
5276	MoUsoCoreWork	GET	304	51.104.136.2:443	https://settings-	US	–	–	whitelisted

	er.exe				win.data.microsoft.com/settings/v3.0/OneSettings/Client?OSVersionFull=10.0.19045.4046.amd64fre.vb_release.191206-1406&LocalDeviceID=s%3ABAD99146-31D3-4EC6-A1A4-BE76F32BA5D4&FlightRing=Retail&AttrDataVer=186&OSUILocale=en-US&OSSkuId=48&App=WOSC&AppVer=&IsFlightingEnabled=0&TelemetryLevel=1&DeviceFamily=Windows.Desktop				
5276	MoUsoCoreWorker.exe	GET	304	51.104.136.2:443	https://settings-win.data.microsoft.com/settings/v3.0/wsd/muse?ProcessorClockSpeed=3094&FlightIds=&UpdateOfferedDays=4294967295&BranchReadinessLevel=CB&OEMManufacturerName=DELL&IsCloudDomainJoined=0&ProcessorIdentifier=AMD64%20Family%2023%20Model%201%20Stepping%202&sku=48&ActivationChannel=Retail&AttrDataVer=186&IsMDMEnrolled=0&ProcessorCores=6&ProcessorModel=AMD%20Ryzen%205%203500%206-Core%20Processor&TotalPhysicalRAM=6144&PrimaryDiskType=4294967295&FlightingBranchName=&ChassisTypeId=1&OEMModelNumber=DELL&SystemVolumeTotalCapacity=260281&sampleId=95271487&deviceClass=Windows.Desktop&App=muse&DisableDualScan=0&AppVer=10.0&OEMSubModel=J5CR&locale=en-US&IsAlwaysOnAlwaysConnectedCapable=0&ms=0&DefaultUserRegion=244&UpdateServiceUrl=http%3A%2F%2Fneverupdatewindows10.com&osVer=10.0.19045.4046.amd64fre.vb_release.191206-1406&os=windows&deviceId=s%3ABAD99146-31D3-4EC6-A1A4-BE76F32BA5D4&DeferQualityUpdatePeriodInDays=0&ring=Retail&DeferFeatureUpdatePeriodInDays=30	US	—	—	whitelisted
6632	SIHClient.exe	GET	304	74.178.76.128:443	https://slscr.update.microsoft.com/SLS/%7B522D76A4-93E1-47F8-B8CE-07C937AD1A1E%7D/x64/10.0.19045.4046/0?CH=686&L=en-US&P=&PT=0x30&WUA=10.0.19041.3996&MK=DELL&MD=DELL	US	—	—	whitelisted
—	—	GET	200	23.11.41.157:80	http://ocsp.digicert.com/MFEwTzBNMEswSTAJBgUrDgMCGgUABBTrjrydRyt%2BApF3GSPypfHBxR5XtQQUs9tIpPmhxdiuNkHMEWNpYim8S8YCEAjTxtAB8my1oj8MfWpz%2F7Y%3D	NL	binary	312 b	whitelisted
6632	SIHClient.exe	GET	200	74.178.240.51:443	https://fe3cr.delivery.mp.microsoft.com/clientwebservice/ping	US	—	—	whitelisted
6632	SIHClient.exe	GET	200	74.178.76.128:443	https://slscr.update.microsoft.com/sls/ping	US	—	—	whitelisted
6632	SIHClient.exe	GET	304	74.178.76.128:443	https://slscr.update.microsoft.com/SLS/%7BE7A50285-D08D-499D-9FF8-180FDC2332BC%7D/x64/10.0.19045.4046/0?CH=686&L=en-US&P=&PT=0x30&WUA=10.0.19041.3996&MK=DELL&MD=DELL	US	—	—	whitelisted
5316	svchost.exe	POST	200	20.190.160.128:443	https://login.live.com/RST2.srf	US	xml	1.24 Kb	whitelisted
—	—	GET	200	204.79.197.203:80	http://oneocsp.microsoft.com/ocsp/MFQwUjBQME4wTDAJBgUrDgMCGgUABBQ3L3%2F%2Fa6ADK8NraY2GXzVaYrHG4AQUb6t%2B2v%2BXQ3LsO2d33oJhNYhHQoUCEzMAAAAGb6JMMcOVb6sAAAAAAAY%3D	US	binary	959 b	whitelisted
7984	svchost.exe	GET	304	40.127.240.158:443	https://settings-win.data.microsoft.com/settings/v3.0/WSD/UpdateHealthTools?os=Windows&osVer=10.0.19041.1.amd64fre.vb_release.191206-&sku=48&deviceClass=Windows.Desktop&locale=en-US&deviceId=s:BAD99146-31D3-4EC6-A1A4-BE76F32BA5D4&sampleId=s:95271487&appVer=10.0.19041.3626&FlightRing=Retail&TelemetryLevel=1&HidOverGattReg=C%3A%5CWINDOWS%5CSystem32%5CDriverStore%5CFileRepository%5Chidbthle.inf_amd64_9610b4821fdf82a5%5CMicrosoft.Bluetooth.Profiles.HidOverGatt.dll&AppVer=&ProcessorIdentifier=AMD64%20Family%2023%20Model%201%20Stepping%202&OEMModel=DELL&UpdateOfferedDays=4294967295&ProcessorManufacturer=AuthenticAMD&InstallDate=1661339444&OEMModelBaseBoard=&BranchReadinessLevel=CB&OEMSubModel=J5CR&IsCloudDomainJoined=0&DeferFeatureUpdatePeriodInDays=30&IsDeviceRetailDemo=0&FlightingBranchName=&OSUILocale=en-US&DeviceFamily=Windows.Desktop&WuClientVer=10.0.19041.3996&UninstallActive=1&IsFlightingEnabled=0&OSSkuId=48&ProcessorClockSpeed=3094&TotalPhysicalRAM=6144&SecureBootCapable=0&App=SedimentPack&ProcessorCores=6&CurrentBranch=vb_release&InstallLanguage=en-US&DeferQualityUpdatePeriodInDays=0&OEMName_Uncleaned=DELL&TPMVersion=0&PrimaryDiskTotalCapacity=262144&InstallationType=Client&AttrDataVer=186&ProcessorModel=AMD%20Ryzen%205%203500%206-Core%20Processor&IsEdgeWithChromiumInstalled=1&OSVersion=10.0.19045.4046&IsMDMEnrolled=0&ActivationChannel=Retail&FirmwareVersion=A.40&TrendInstalledKey=1&OSArchitecture=AMD64&DefaultUserRegion=244&UpdateManagementGroup=2	US	—	—	whitelisted
5316	svchost.exe	POST	400	20.190.160.128:443	https://login.live.com/ppsecure/deviceaddcredential.srf	US	text	203 b	whitelisted
5316	svchost.exe	POST	400	20.190.160.128:443	https://login.live.com/ppsecure/deviceaddcredential.srf	US	text	203 b	whitelisted
5316	svchost.exe	GET	200	23.11.41.157:80	http://ocsp.digicert.com/MFEwTzBNMEswSTAJBgUrDgMCGgUABBQ50otx%2Fh0Ztl%2Bz8SiPI7wEWVxDlQQUTiJUIBiV5uNu5g%2F6%2BrkS7QYXjzkCEAz1vQYrVgL0erhQLCPM8GY%3D	NL	binary	471 b	whitelisted
5316	svchost.exe	POST	400	20.190.160.128:443	https://login.live.com/ppsecure/deviceaddcredential.srf	US	text	203 b	whitelisted
5316	svchost.exe	POST	400	20.190.160.128:443	https://login.live.com/ppsecure/deviceaddcredential.srf	US	text	203 b	whitelisted
7984	svchost.exe	GET	200	23.216.77.10:80	http://crl.microsoft.com/pki/crl/products/MicRooCerAut2011_2011_03_22.crl	NL	binary	825 b	whitelisted
7984	svchost.exe	GET	200	51.124.78.146:443	https://settings-win.data.microsoft.com/settings/v3.0/WSD/WaaSAssessment?os=Windows&osVer=10.0.19041.1.amd64fre.vb_release.191206-&ring=Retail&sku=48&deviceClass=Windows.Desktop&locale=en-US&deviceId=BAD99146-31D3-4EC6-A1A4-BE76F32BA5D4&FlightRing=Retail&TelemetryLevel=1&HidOverGattReg=C%3A%5CWINDOWS%5CSystem32%5CDriverStore%5CFileRepository%5Chidbthle.inf_amd64_9610b4821fdf82a5%5CMicrosoft.Bluetooth.Profiles.HidOverGatt.dll&AppVer=10.0&ProcessorIdentifier=AMD64%20Family%2023%20Model%201%20Stepping%202&OEMModel=DELL&UpdateOfferedDays=4294967295&ProcessorManufacturer=AuthenticAMD&InstallDate=1661339444&OEMModelBaseBoard=&BranchReadinessLevel=CB&OEMSubModel=J5CR&IsCloudDomainJoined=0&DeferFeatureUpdatePeriodInDays=30&IsDeviceRetailDemo=0&FlightingBranchName=&OSUILocale=en-US&DeviceFamily=Windows.Desktop&WuClientVer=10.0.19041.399	US	text	5.74 Kb	whitelisted

					6&UninstallActive=1&IsFlightingEnabled=0&OSSkuId=48&ProcessorClockSpeed=3094&TotalPhysicalRAM=6144&SecureBootCapable=0&App=WaaSAssessment&ProcessorCores=6&CurrentBranch=vb_release&InstallLanguage=en-US&DeferQualityUpdatePeriodInDays=0&ServicingBranch=CB&OEMName_Uncleaned=DELL&TPMVersion=0&PrimaryDiskTotalCapacity=262144&InstallationType=Client&AttrDataVer=186&ProcessorModel=AMD%20Ryzen%205%203500%206-Core%20Processor&IsEdgeWithChromiumInstalled=1&OSVersion=10.0.19045.4046&IsMDMEnrolled=0&ActivationChannel=Retail&HonorWUfBDeferrals=0&FirmwareVersion=A.40&TrendInstalledKey=1&OSArchitecture=AMD64&DefaultUserRegion=244&UpdateManagementGroup=2				
7984	svchost.exe	GET	200	88.221.169.152:80	http://www.microsoft.com/pkiops/crl/MicSecSerCA2011_2011-10-18.crl	US	binary	814 b	whitelisted
5316	svchost.exe	POST	400	20.190.160.128:443	https://login.live.com/ppsecure/deviceaddcredential.srf	US	text	203 b	whitelisted
5316	svchost.exe	POST	400	20.190.160.128:443	https://login.live.com/ppsecure/deviceaddcredential.srf	US	text	203 b	whitelisted
5316	svchost.exe	POST	400	20.190.160.128:443	https://login.live.com/ppsecure/deviceaddcredential.srf	US	text	203 b	whitelisted
5316	svchost.exe	POST	400	20.190.160.128:443	https://login.live.com/ppsecure/deviceaddcredential.srf	US	text	203 b	whitelisted
5276	MoUsoCoreWorker.exe	GET	200	51.104.136.2:443	https://settings-win.data.microsoft.com/settings/v3.0/FlightSettings/FSService?ProcessorClockSpeed=3094&IsRetailOS=1&OEMManufacturerName=DELL&FlightingPolicyValue=3&EnablePreviewBuilds=4294967295&OSVersionFull=10.0.19045.4046.amd64fre.vb_release.191206-1406&ManagePreviewBuilds=3&BranchReadinessLevelSource=0&AttrDataVer=186&ProcessorCores=6&BranchReadinessLevelRaw=16&TotalPhysicalRAM=6144&TPMVersion=0&OEMModelNumber=DELL&SystemVolumeTotalCapacity=260281&DeviceId=s%3ABAD99146-31D3-4EC6-A1A4-BE76F32BA5D4&App=FSS&AppVer=10.0&SmartActiveHoursState=1&ActiveHoursStart=20&SecureBootCapable=0&ActiveHoursEnd=13&DeviceFamily=Windows.Desktop	US	text	87.3 Kb	whitelisted
5316	svchost.exe	POST	400	20.190.160.128:443	https://login.live.com/ppsecure/deviceaddcredential.srf	US	text	203 b	whitelisted
5316	svchost.exe	POST	400	20.190.160.128:443	https://login.live.com/ppsecure/deviceaddcredential.srf	US	text	203 b	whitelisted
5316	svchost.exe	POST	400	20.190.160.128:443	https://login.live.com/ppsecure/deviceaddcredential.srf	US	text	203 b	whitelisted
6632	SIHClient.exe	GET	200	88.221.169.152:80	http://www.microsoft.com/pkiops/crl/Microsoft%20ECC%20Product%20Root%20Certificate%20Authority%202018.crl	US	binary	419 b	whitelisted
6632	SIHClient.exe	GET	200	88.221.169.152:80	http://www.microsoft.com/pkiops/crl/Microsoft%20ECC%20Update%20Secure%20Server%20CA%202.1.crl	US	binary	407 b	whitelisted
5316	svchost.exe	POST	400	20.190.160.128:443	https://login.live.com/ppsecure/deviceaddcredential.srf	US	text	203 b	whitelisted

Connections

PID	Process	IP	Domain	ASN	CN	Reputation
4	System	192.168.100.255:137	—	Not routed	—	whitelisted
7984	svchost.exe	51.124.78.146:443	settings-win.data.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	whitelisted
5276	MoUsoCoreWorker.exe	51.124.78.146:443	settings-win.data.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	whitelisted
—	—	48.192.1.65:443	activation-v2.sls.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	whitelisted
—	—	2.16.241.217:443	www.bing.com	AKAMAI-ASN1	NL	whitelisted
—	—	23.11.41.157:80	ocsp.digicert.com	AKAMAI-AMS	NL	whitelisted
—	—	204.79.197.203:80	oneocsp.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	whitelisted
4	System	192.168.100.255:138	—	Not routed	—	whitelisted
5316	svchost.exe	20.190.160.128:443	login.live.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	whitelisted
3428	svchost.exe	172.211.123.249:443	client.wns.windows.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	whitelisted
5316	svchost.exe	23.11.41.157:80	ocsp.digicert.com	AKAMAI-AMS	NL	whitelisted
7984	svchost.exe	23.216.77.10:80	crl.microsoft.com	AKAMAI-ASN1	NL	whitelisted
7984	svchost.exe	88.221.169.152:80	www.microsoft.com	AKAMAI-AS	US	whitelisted
5276	MoUsoCoreWorker.exe	51.104.136.2:443	settings-win.data.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	whitelisted
6632	SIHClient.exe	74.178.76.128:443	slscr.update.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	whitelisted
6632	SIHClient.exe	88.221.169.152:80	www.microsoft.com	AKAMAI-AS	US	whitelisted
6632	SIHClient.exe	74.178.240.51:443	fe3cr.delivery.mp.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	whitelisted
7984	svchost.exe	40.127.240.158:443	settings-win.data.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	whitelisted

DNS requests

Domain	IP	Reputation
settings-win.data.microsoft.com	51.124.78.146 51.104.136.2 40.127.240.158	whitelisted
activation-v2.sls.microsoft.com	48.192.1.65	whitelisted
www.bing.com	2.16.241.217 2.16.241.208 2.16.241.223 2.16.241.200 2.16.241.207 2.16.241.220 2.16.241.222 2.16.241.219 2.16.241.206	whitelisted
google.com	192.178.183.139 192.178.183.138 192.178.183.100 192.178.183.102 192.178.183.113 192.178.183.101	whitelisted
ocsp.digicert.com	23.11.41.157	whitelisted
oneocsp.microsoft.com	204.79.197.203	whitelisted
login.live.com	20.190.160.128 40.126.32.140 20.190.160.22 20.190.160.131 40.126.32.134 20.190.160.65 20.190.160.17 40.126.32.136	whitelisted
client.wns.windows.com	172.211.123.249	whitelisted
crl.microsoft.com	23.216.77.10 23.216.77.15 23.216.77.23	whitelisted
www.microsoft.com	88.221.169.152	whitelisted
slscr.update.microsoft.com	74.178.76.128	whitelisted
fe3cr.delivery.mp.microsoft.com	74.178.240.51	whitelisted
self.events.data.microsoft.com	52.168.112.67	whitelisted

Threats

PID	Process	Class	Message
7984	svchost.exe	Unknown Traffic	ET USER_AGENTS Microsoft Dr Watson User-Agent (MSDW)

Debug output strings

No debug info

ANY.RUN INTERACTIVE MALWARE ANALYSIS
Interactive malware hunting service ANY.RUN
© 2017-2026 ANY.RUN ALL RIGHTS RESERVED

Part B: In class malware analysis of dontbother.exe

Files as part of this lab setup:

This lab uses a assignment zip:

Lab Setup

Step1: Download malware from the link below:

1) https://drive.google.com/drive/folders/1LDCuWu-6gFGtZiIYluGd8BHDaURegvOo
2) Install REMnux

Part 1: Static Analysis

Step2: Unzip the file:

3) Command: Run 7z x assignment.7z
4) Password: pleasebesafe

```
remnux@remnux:~/Downloads$ 7z x assignment.7z

7-Zip 23.01 (x64) : Copyright (c) 1999-2023 Igor Pavlov : 2023-06-20
 64-bit locale=C.UTF-8 Threads:2 OPEN_MAX:1024

Scanning the drive for archives:
1 file, 140138 bytes (137 KiB)

Extracting archive: assignment.7z
--
Path = assignment.7z
Type = 7z
Physical Size = 140138
Headers Size = 170
Method = LZMA2:192k BCJ 7zAES
Solid = -
Blocks = 1


Enter password (will not be echoed):
Everything is Ok

Size:       189337
Compressed: 140138
remnux@remnux:~/Downloads$ ls
README.gif  assignment.7z  dontbother.exe  malware.7z  satan.exe
remnux@remnux:~/Downloads$
```

Step6: Inspect satan.exe:

5) Command: file dontbother.exe

```
remnux@remnux:~/Downloads$ file dontbother.exe
dontbother.exe: PE32 executable (GUI) Intel 80386, for MS Windows, 4 sections
remnux@remnux:~/Downloads$
```

Observation:

- PE32 (32-bit) Windows executable, confirmed via magic bytes
- GUI subsystem: likely to suppress console window visibility
- x86 (Intel 80386) architecture, compatible with 32/64-bit Windows via WOW64
- Only 4 PE sections detected: low count, possible packing or obfuscation

Step3: find the hash of the malware - hashes are used to identify malwares:

6) Command: sha256sum dontbother.exe

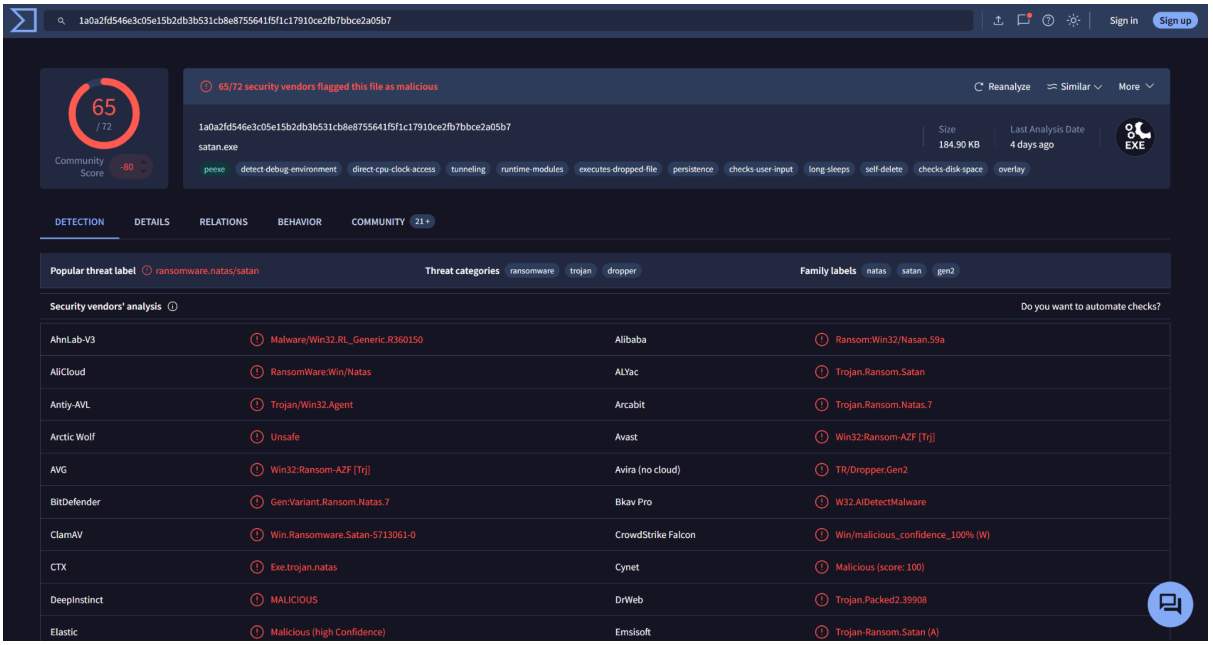
```
remnux@remnux:~/Downloads$ sha256sum dontbother.exe
1a0a2fd546e3c05e15b2db3b531cb8e8755641f5f1c17910ce2fb7bbce2a05b7  dontbother.exe
```

Sha256:
1a0a2fd546e3c05e15b2db3b531cb8e8755641f5f1c17910ce2fb7bbce2a05b7

```
remnux@remnux:~/Downloads$ md5sum dontbother.exe
c9c341eaf04c89933ed28cbc2739d325  dontbother.exe
remnux@remnux:~/Downloads$
```

Md5: c9c341eaf04c89933ed28cbc2739d325

Step4: Perform a VirusTotal file hash lookup:

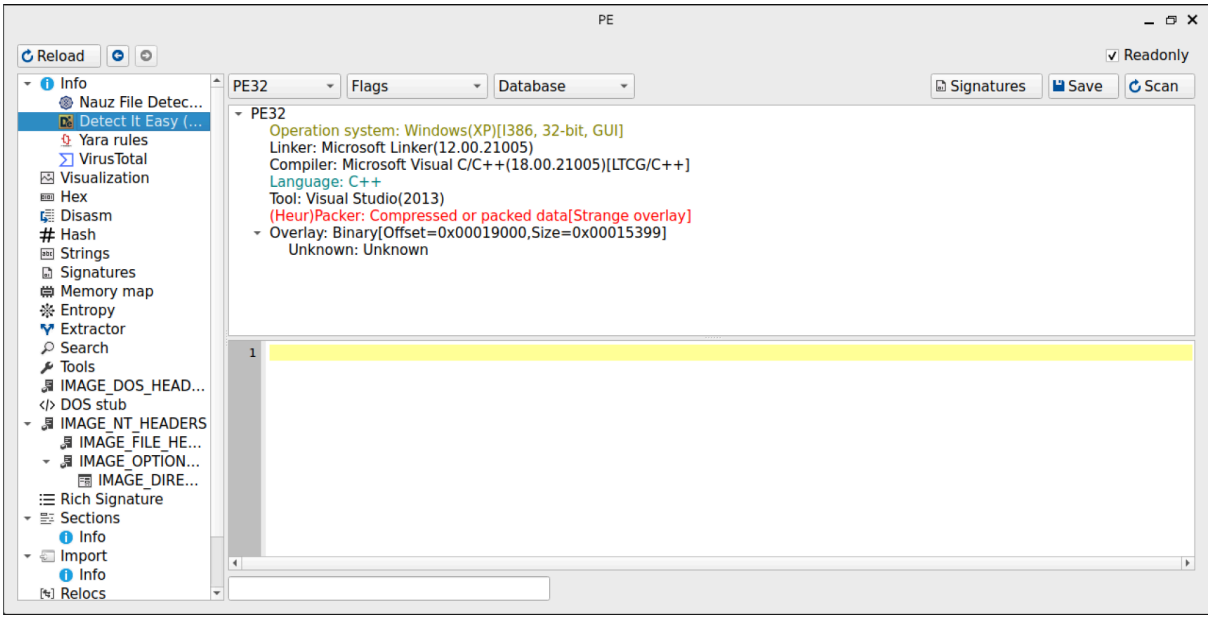


Explanation:
From the above we see that 65 out of 72 vendors flagged the sample as malicious with a community score of -80, confirming it as high-confidence ransomware. It is consistently identified as the Satan/Natas ransomware family (gen2) and classified as a trojan dropper. Behavioral tags indicate sandbox evasion via debug environment detection and long sleeps, persistence mechanisms, self-deletion

post-execution, and possible C2 tunneling. The packed signature noted by DrWeb aligns with the low section count observed in initial triage.

Step5: deep inspect the file with die (detect it easy):

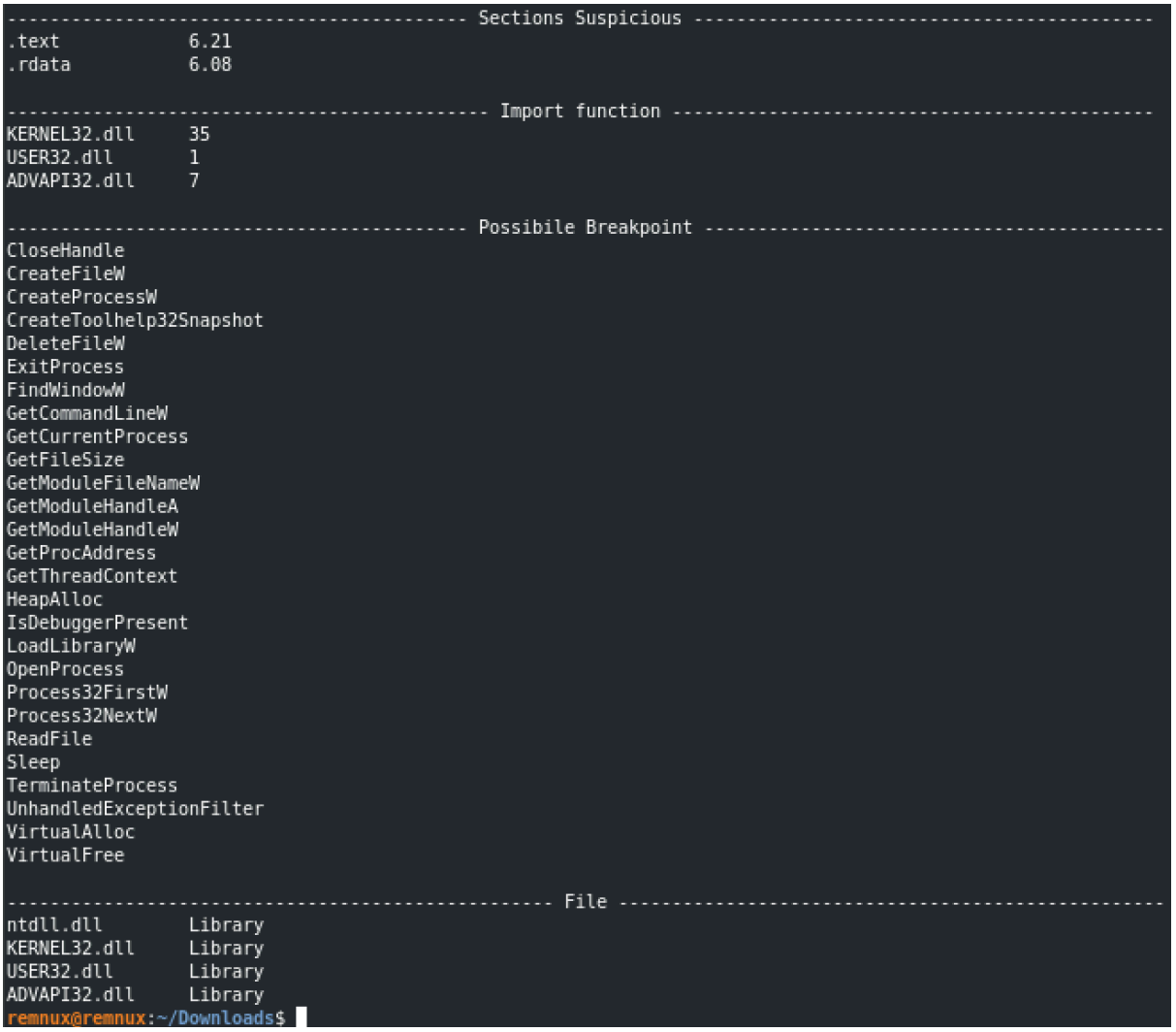
7) Command: die dontbother.exe



Step6: linux utility for file information:

8) Command: peframe dontbother.exe

```
remnux@remnux:~/Downloads$ peframe dontbother.exe

------------------------------- File Information (time: 0:00:02.143386) -------------------------------
filename          dontbother.exe
filetype          PE32 executable (GUI) Intel 80386, for MS Windows, 4 sections
filesize          189337
md5               c9c341eaf04c89933ed28cbc2739d325
sha1              c5b7d47aef3bd33a24293138fcba3a5ff286c2a8
sha256            1a0a2fd546e3c05e15b2db3b531cb8e8755641f5f1c17910ce2fb7bbce2a05b7
virustotal        X-Apikey header is missing
imagebase         0x400000
entrypoint        0x13b9
imphash           65e9607e6f28a7852bb41a6e2e439a92
datetime          2017-02-23 19:28:24
dll               False
directories       import, tls, relocations
sections          .data, .reloc, .text *, .rdata *
features          antidbg, crypto

---------------------------------------------- Yara Plugins ----------------------------------------------
IsPE32
IsWindowsGUI
IsPacked
HasOverlay
HasRichSignature
Advapi Hash API
CRC32 poly Constant

------------------------------------------------ Behavior ------------------------------------------------
SEH  vectored
anti dbg
Xor

-------------------------------------------------- Crypto --------------------------------------------------
Advapi Hash API
CRC32 poly Constant

------------------------------------------------ Anti Debug ------------------------------------------------
FindWindowW
IsDebuggerPresent
IsProcessorFeaturePresent
Process32FirstW
Process32NextW
TerminateProcess
UnhandledExceptionFilter
```

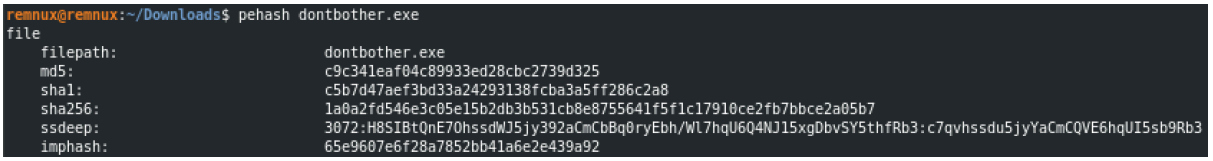


Observation and Explanation:

- peframe confirms the binary is a packed PE32 GUI executable compiled on 2017-02-23, with high-entropy sections (.text at 6.21, .rdata at 6.08) strongly indicating packed or encrypted code.
- YARA rules flag it as IsPacked, HasOverlay, and HasRichSignature.
- Detected features include antidbg and crypto, with XOR-based obfuscation and CRC32/Advapi Hash API usage for cryptographic operations.
- The anti-debug layer employs IsDebuggerPresent, FindWindowW, UnhandledExceptionFilter, and process enumeration via Process32FirstW/NextW to detect and evade analysis environments.
- Import analysis shows 35 KERNEL32, 7 ADVAPI32, and 1 USER32 imports - notable calls include VirtualAlloc, LoadLibraryW, GetProcAddress (classic shellcode/unpacking pattern), CreateProcessW, DeleteFileW (self-delete), and OpenProcess (process injection candidate).
- The presence of nss.cM as an unknown file reference is also suspicious and warrants further investigation.

Step7: linux utility for file information:

9) Command: pehash dontbother.exe



Step8: Check for strings. Strings won't work as malware code is generally in different encodings. We use floss instead:

10)Command: floss satan.exe

```
| FLOSS STATIC STRINGS: UTF-16LE (3) |
+------------------------------------+

Microsoft Enhanced Cryptographic Provider v1.0
Microsoft Enhanced RSA and AES Cryptographic Provider (Prototype)
Microsoft Enhanced RSA and AES Cryptographic Provider


 ___________________________
  FLOSS STACK STRINGS (0)
 ___________________________




 ___________________________
  FLOSS TIGHT STRINGS (2)
 ___________________________

P2AD
y80o


 ____________________________
  FLOSS DECODED STRINGS (107)
 ____________________________

netmon.exe
WinDbgFrameClass
\SAMPLE
filemon.exe
VIRUS
SbieDll.dll
ObsidianGUI
PETools.exe
ntdll.dll
SANDBOX
procmon.exe
dbghelp.dll
LordPE.exe
avghookx.dll
C:\InsideTm
snxhk.dll
procexp.exe
SysInspector.exe
Immunity Debugger
MALWARE
```

Observation: Malware is heavily encrypted. Could indicate a ransomware

```
MALWARE
avghooka.dll
api_log.dll
idaq.exe
proc_analyser.exe
MALTEST
dir_watch.dll
idaq64.exe
TEQUILABOOMBOOM
sysAnalyzer.exe
vmcheck.dll
ollydbg.exe
devenv.exe
ImmunityDebugger.exe
kernel32.dll
Zeta Debugger
sample.ex
wpespy.dll
sniff_hit.exe
ProcessHacker.exe
Wireshark.exe
pstorec.dll
windbg.exe
tcpview.exe
OLLYDBG
dwmapi.dll
user32.dll
ntdll.d
csrss.e
dumpcap.exe
joeboxcontrol.exe
autoruns.exe
Rock Debugger
HookExplorer.exe
joeboxserver.exe
autorunsc.exe
ImportREC.exe
NtGetContextThread
LdrLoadDll
NtSetContextThread
NtUnmapViewOfSection
NtResumeThread
NtClose
ntdll.dll
NtQueryInformationProcess
kernel32.dll
```

```
kernel32.dll
VirtualAllocEx
CreateProces
IsDebuggerPresent
CheckRemoteDebuggerPresent
poAe
NtTerminateProcess
BlockInput
NtReadVirtualMemory
PuAt
wine_get_unix_file_name
NtWriteVirtualMemory
dir_watch.dll
vmcheck.dll
dbghelp.dll
SbieDll.dll
pstorec.dll
api_log.dll
wpespy.dll
snxhk.dll
ImmunityDebugger.exe
joeboxcontrol.exe
ProcessHacker.exe
proc_analyser.exe
HookExplorer.exe
SysInspector.exe
sysAnalyzer.exe
ImportREC.exe
Wireshark.exe
autorunsc.exe
sniff_hit.exe
autoruns.exe
ollydbg.exe
filemon.exe
PETools.exe
procmon.exe
dumpcap.exe
procexp.exe
tcpview.exe
idaq64.exe
windbg.exe
LordPE.exe
idaq.exe

remnux@remnux:~/Downloads$
```

Observation and Explanation:
Anti-Analysis Tool Detection:
The decoded strings reveal an extensive hardcoded blacklist of analysis tools the malware actively checks for, including debuggers (OllyDbg, WinDbg, ImmunityDebugger, x64dbg), process monitors (procmon.exe, procexp.exe, ProcessHacker.exe), network tools (Wireshark, tcpview, dumpcap), sandbox indicators (vmcheck.dll, SbieDll.dll, dir_watch.dll), and RE tools (PETools, ImportREC, LordPE). Keywords like MALWARE, MALTEST, VIRUS, SANDBOX, and \SAMPLE are explicitly checked, suggesting the malware scans its own execution path or environment strings to detect sandbox deployments.

Crypto Providers:
FLOSS UTF-16LE strings surface three Microsoft cryptographic provider references: Enhanced Cryptographic Provider v1.0, Enhanced RSA and AES Provider, and its prototype variant. This directly corroborates the ADVAPI32 Crypt* import chain seen in PE Studio and confirms AES/RSA usage, consistent with ransomware file encryption.

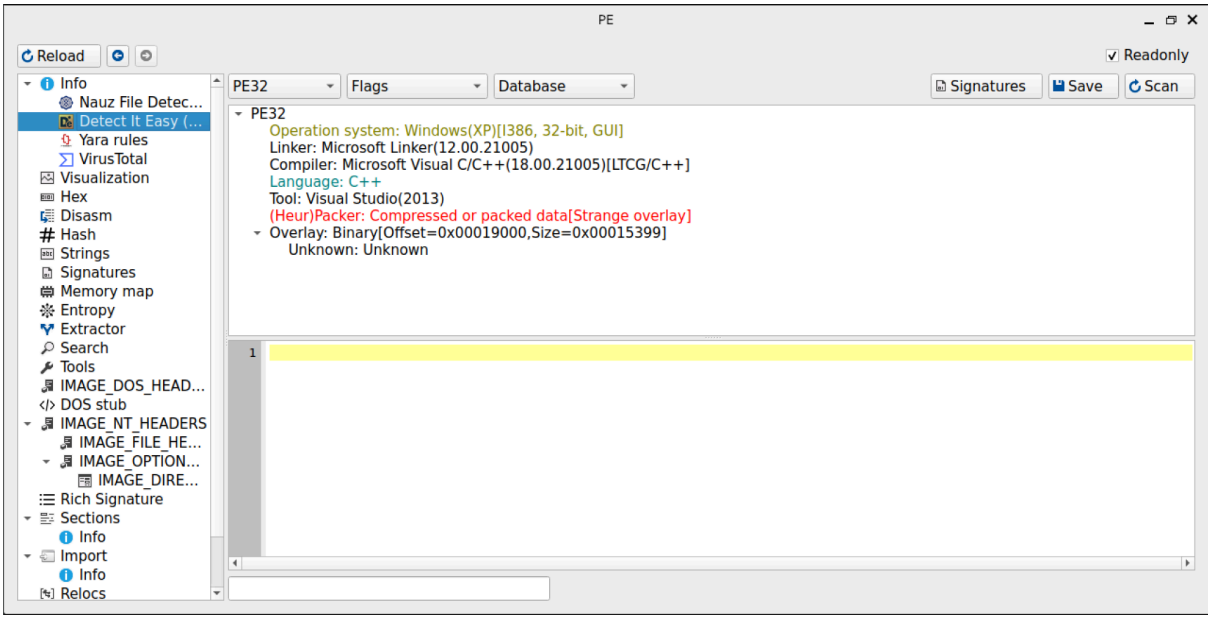
NT-Level Evasion:
Decoded strings include direct NT API calls: NtGetContextThread, NtSetContextThread, NtUnmapViewOfSection, NtResumeThread, NtWriteVirtualMemory, NtReadVirtualMemory, NtQueryInformationProcess and LdrLoadDll. This indicates process hollowing or injection capability operating below the standard Win32 API layer to evade usermode hooks.

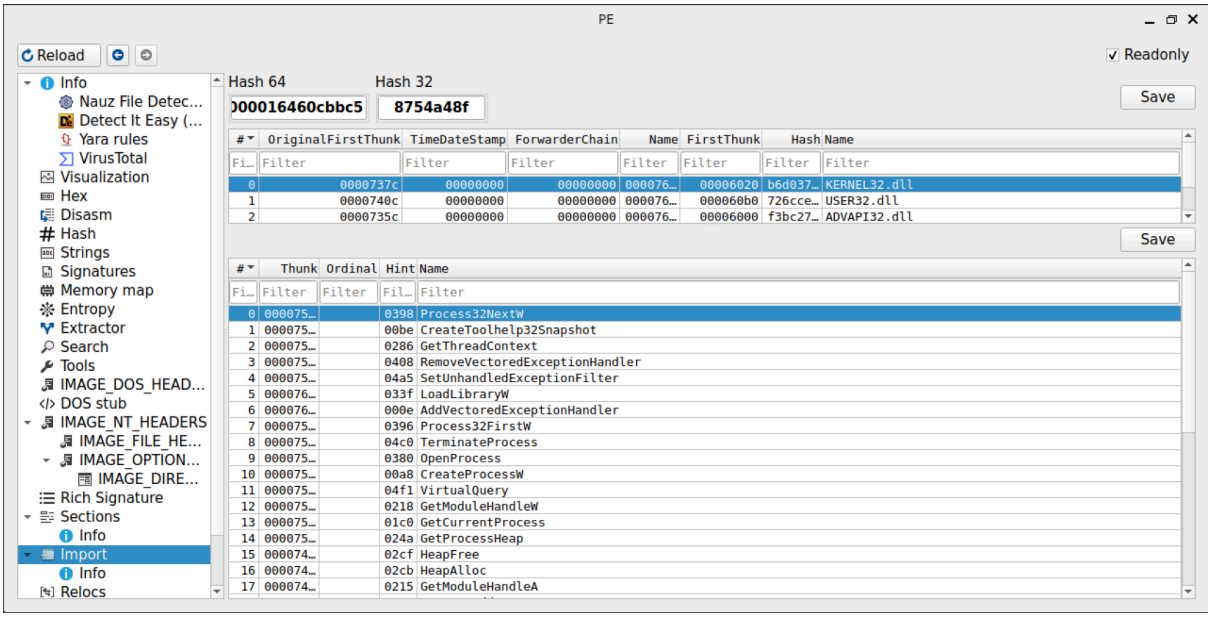
Tight Strings:
Only two tight strings decoded: P2AD and y80o, likely obfuscated keys or internal markers used during runtime decryption.

Step9: Find imports using die:

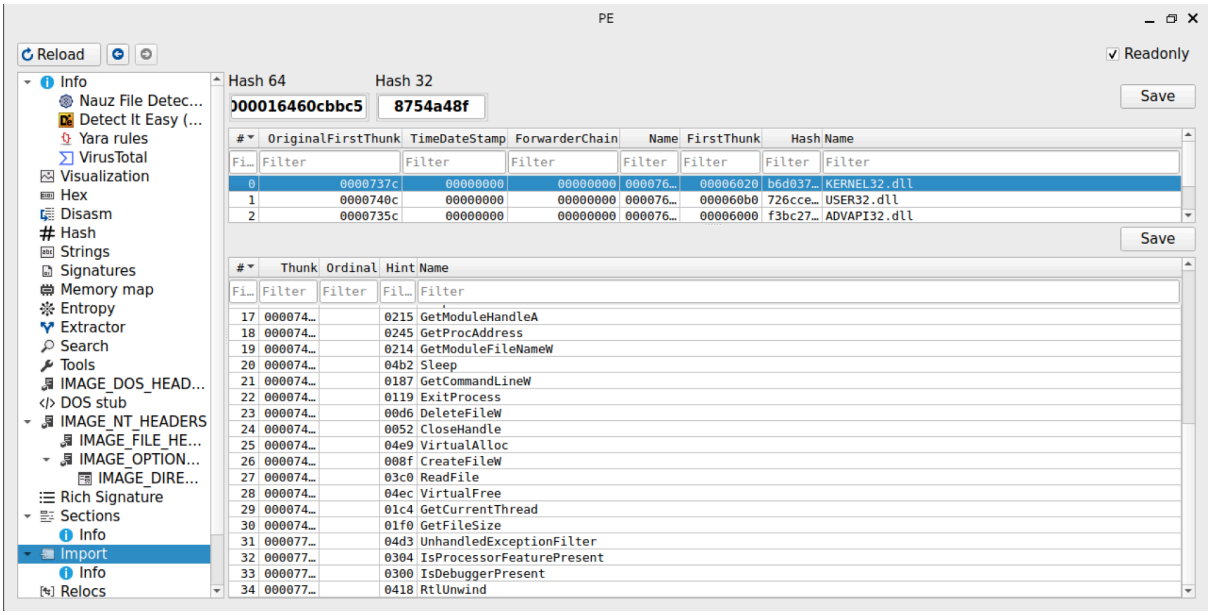
11) Command: die satan.exe

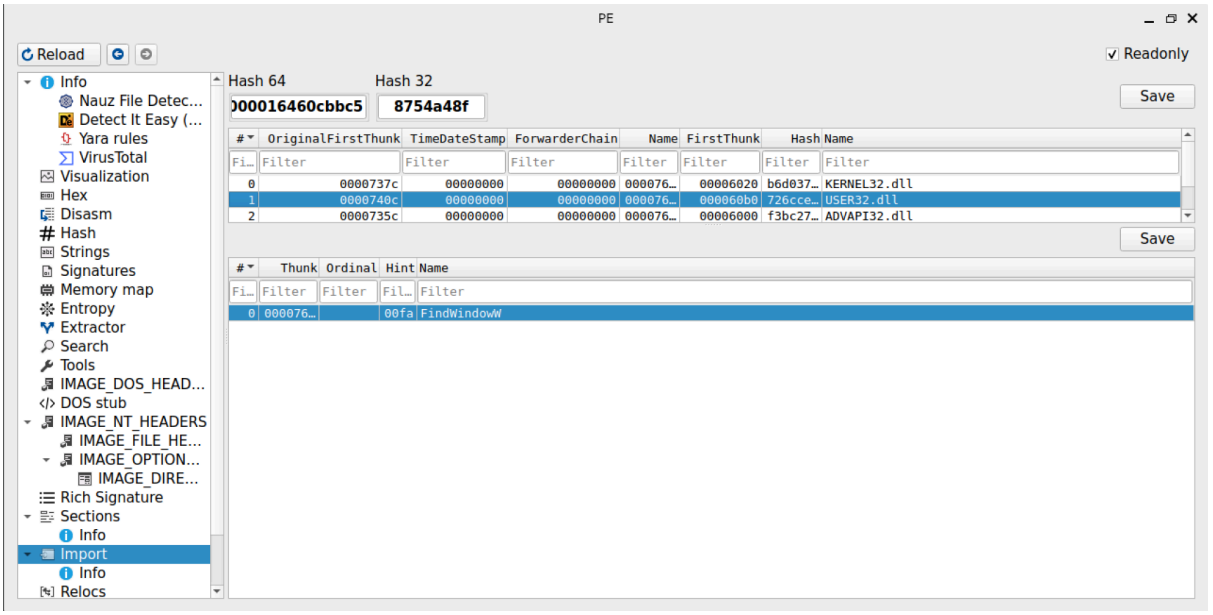
Jan - May 2026 Assignment SUBMISSION_UE23CS343BB6

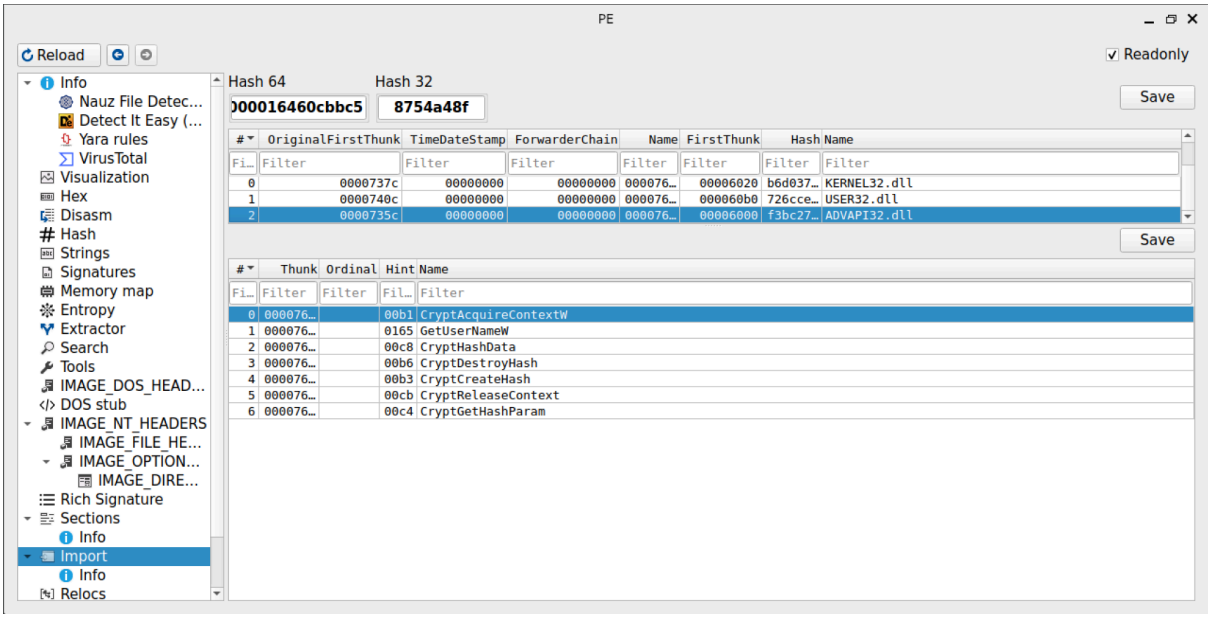




Jan - May 2026 Assignment SUBMISSION_UE23CS343BB6







Observation and Explanation:

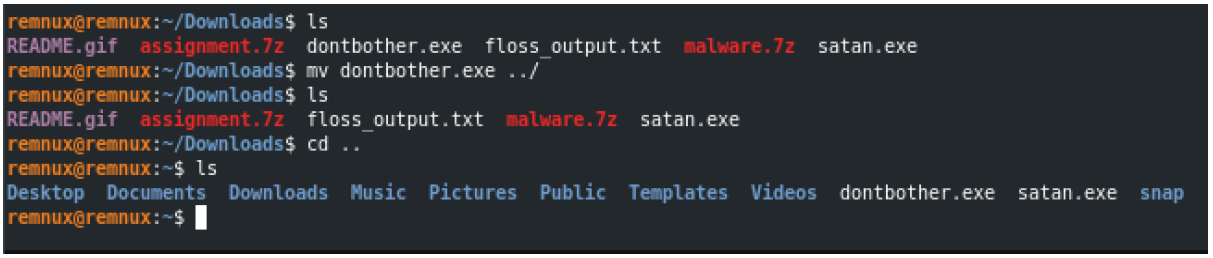
- DIE identifies the binary as compiled with Microsoft Visual C++ 18.00 (Visual Studio 2013) using LTCG, targeting Windows XP x86 GUI. It heuristically flags the binary as a packer with compressed data and a strange overlay at offset 0x00019000 (size 0x00015399), consistent with the packed payload observed in earlier analysis.
- The import table reveals three DLLs: KERNEL32, USER32, and ADVAPI32. KERNEL32 imports confirm process manipulation (CreateProcessW, OpenProcess, TerminateProcess), memory operations (VirtualAlloc, VirtualFree, HeapAlloc), file operations (CreateFileW, ReadFile, DeleteFileW), and anti-debug checks (IsDebuggerPresent, UnhandledExceptionFilter).
- The vectored exception handler imports (AddVectoredExceptionHandler, RemoveVectoredExceptionHandler) alongside GetThreadContext suggest runtime unpacking or anti-analysis trampolining.
- ADVAPI32 imports are entirely crypto-related: CryptAcquireContextW, CryptCreateHash, CryptHashData, CryptGetHashParam, CryptDestroyHash, CryptReleaseContext, a complete hashing pipeline consistent with file encryption for ransomware operation.
- USER32 contributes only FindWindowW, used for debugger and analysis tool window detection.

Part 2: Dynamic Analysis

(IMPORTANT): Create a snapshot of vm as after dynamic analysis, the vm will be corrupted:

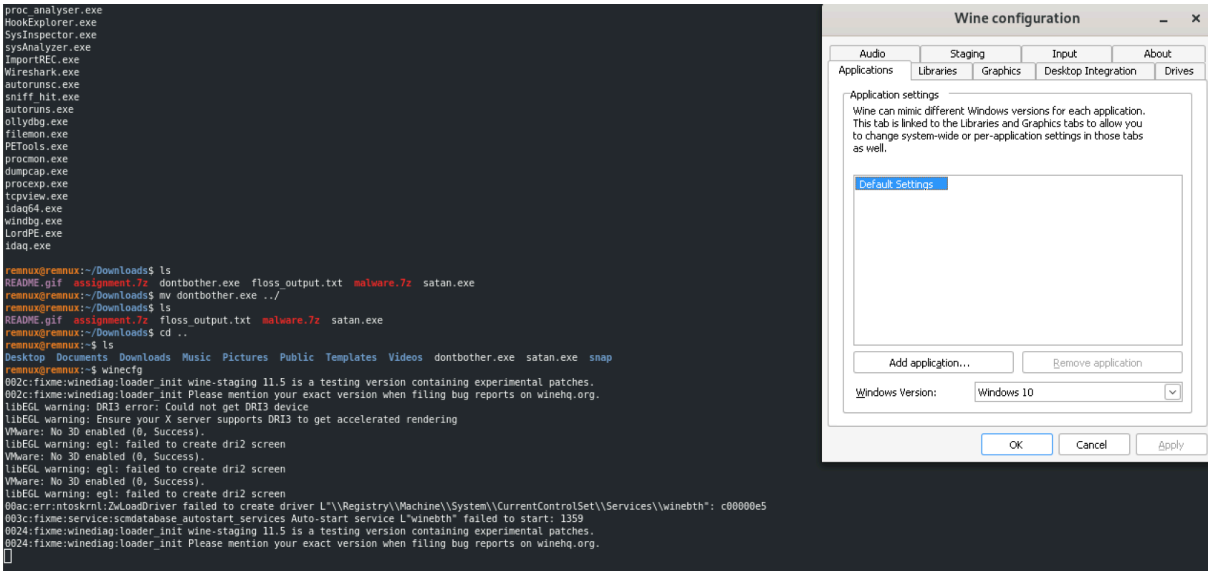
Step10: Move satan.exe to upper dir

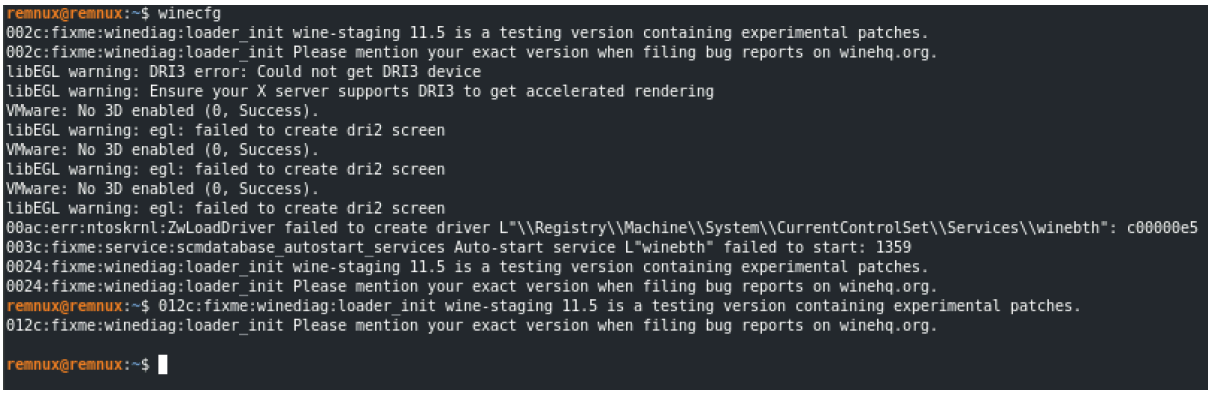
12)Command: mv dontbother.exe ../



Step11: Setup wine:

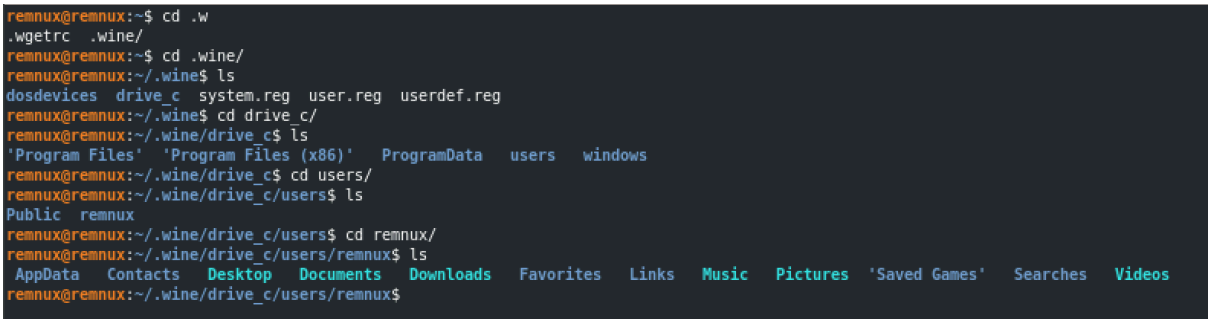
13)Command: winecfg





Explanation:
Wine is setup on the remnux vm to run the file for dynamic analysis. As we can see above wine is successfully setup.

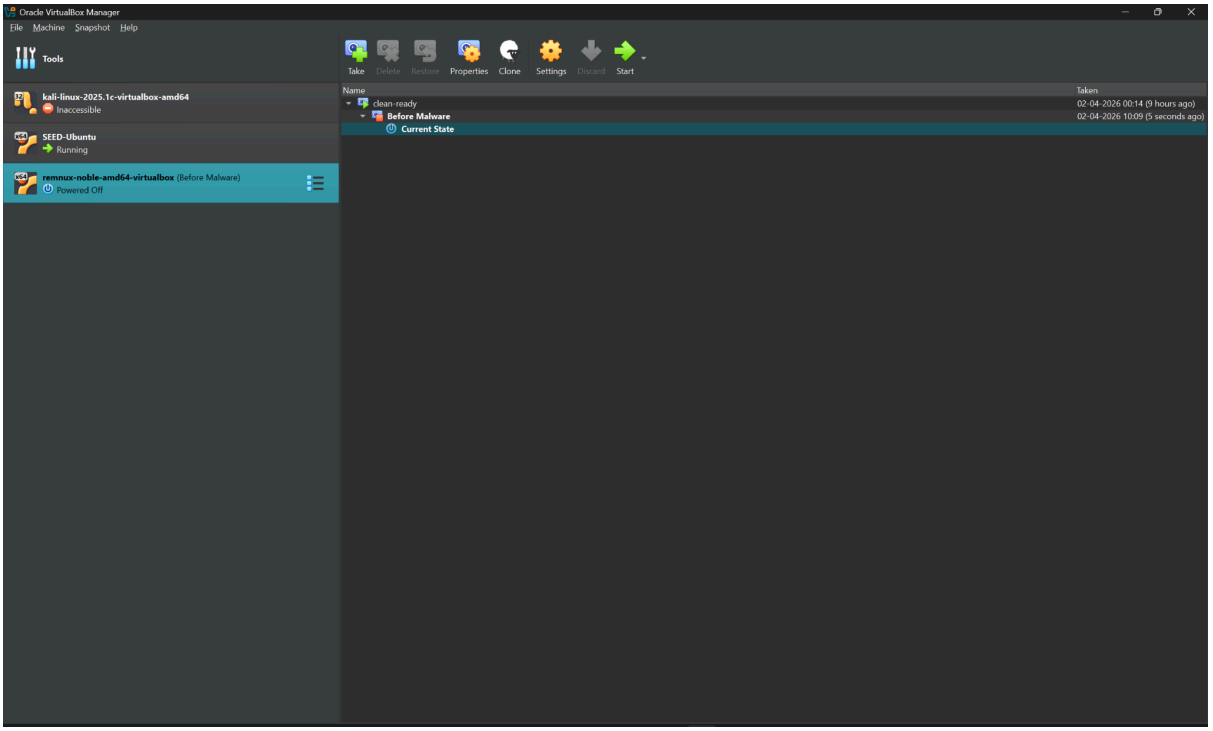
Step12: Verify setup:



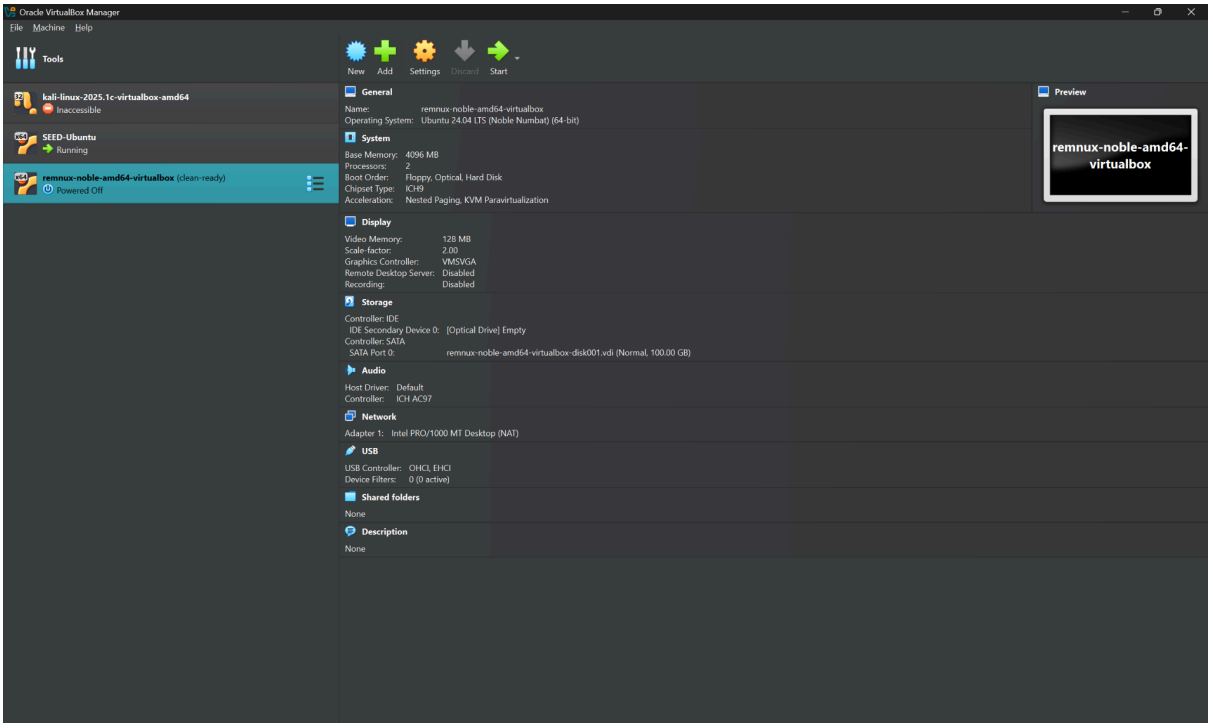
Step13: Power the vm off:

14)Command: poweroff

Step16: Take the snapshot:



Step14: See the snapshot details:



Step15: Install inotifywait:

15)Command: sudo apt install inotifywait-tools

```
remnux@remnux:~$ sudo apt install inotify-tools
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
inotify-tools is already the newest version (3.22.6.0-4).
0 upgraded, 0 newly installed, 0 to remove and 29 not upgraded.
```

Step16: Setup a directory watch:

16)Command: inotifywait -m -r ~/.wine --format '%T %e %w%f' --timefmt '%H:%M:%S' -e create,modify,delete,move

```
remnux@remnux:~$ inotifywait -m -r ~/.wine --format '%T %e %w%f' --timefmt '%H:%M:%S' -e create,modi
fy,delete,move
Setting up watches.  Beware: since -r was given, this may take a while!
Watches established.
```

Step17: run tcpdump:

17)Command: sudo tcpdump -i any -n -w dontbother_capture.pcap 'not port 22'

```
remnux@remnux:~$ sudo tcpdump -i any -n -w dontbother_capture.pcap 'not port 22'
tcpdump: data link type LINUX_SLL2
tcpdump: listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
```

Step18: Run wine:

18)Command: wine dontbother.exe

```
remnux@remnux:~$ wine dontbother.exe
002c:fixme:winediag:loader_init wine-staging 11.5 is a testing version containing experimental patches.
002c:fixme:winediag:loader_init Please mention your exact version when filing bug reports on winehq.org.
libEGL warning: DRI3 error: Could not get DRI3 device
libEGL warning: Ensure your X server supports DRI3 to get accelerated rendering
VMware: No 3D enabled (0, Success).
libEGL warning: egl: failed to create dri2 screen
VMware: No 3D enabled (0, Success).
libEGL warning: egl: failed to create dri2 screen
VMware: No 3D enabled (0, Success).
libEGL warning: egl: failed to create dri2 screen
00b0:err:ntoskrnl:ZwLoadDriver failed to create driver L"\\Registry\\Machine\\System\\CurrentControlSet\\Services\\winebth": c00000e5
003c:fixme:service:scmdatabase_autostart_services Auto-start service L"winebth" failed to start: 1359
010c:fixme:winediag:loader_init wine-staging 11.5 is a testing version containing experimental patches.
010c:fixme:winediag:loader_init Please mention your exact version when filing bug reports on winehq.org.
0024:fixme:keyboard:BlockInput (1): stub
remnux@remnux:~$ 0120:fixme:volume:GetVolumeNameForVolumeMountPointW Mounted Folders are not yet supported
0128:fixme:keyboard:BlockInput (1): stub
0140:fixme:volume:GetVolumeNameForVolumeMountPointW Mounted Folders are not yet supported
remnux@remnux:~$
```

Explanation:

- BlockInput called but stubbed by Wine, the malware attempted to block user input, a common ransomware tactic to prevent interference during encryption
- GetVolumeNameForVolumeMountPointW called twice, scanning mounted volumes, consistent with ransomware enumerating drives before encryption
- Driver creation attempt via ZwLoadDriver failed (c00000e5), likely attempted to load a kernel-level component, blocked by Wine's limitations

inotifywait watch:

```
remnux@remnux:~$ inotifywait -m -r ~/.wine --format "%T %e %w%f" --timefmt "%H:%M:%S" -e create,modify,delete,move
Setting up watches.  Beware: since -r was given, this may take a while!
Watches established.
10:02:56 CREATE,ISDIR /home/remnux/.wine/drive_c/users/remnux/AppData/Roaming/Esbo
10:02:56 MODIFY /home/remnux/.wine/drive_c/users/remnux/AppData/Roaming/Esbo/uzico.exe
10:02:56 MODIFY /home/remnux/.wine/drive_c/users/remnux/AppData/Roaming/Esbo/uzico.exe
10:02:56 CREATE /home/remnux/.wine/drive_c/users/remnux/AppData/Local/Temp/tmp_65a9ca40.bat
10:02:56 MODIFY /home/remnux/.wine/drive_c/users/remnux/AppData/Local/Temp/tmp_65a9ca40.bat
10:02:56 MODIFY /home/remnux/.wine/drive_c/users/remnux/AppData/Local/Temp/tmp_65a9ca40.bat
10:02:56 MODIFY /home/remnux/.wine/drive_c/users/remnux/AppData/Local/Temp/tmp_65a9ca40.bat
10:02:57 DELETE /home/remnux/.wine/drive_c/users/remnux/AppData/Local/Temp/tmp_65a9ca40.bat
10:03:25 CREATE /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY
```

```
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MOVED_FROM /home/remnux/.wine/reg14770000.tmp
10:03:25 MOVED_TO /home/remnux/.wine/system.reg
10:03:25 CREATE /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MOVED_FROM /home/remnux/.wine/reg14770000.tmp
10:03:25 MOVED_TO /home/remnux/.wine/userdef.reg
10:03:25 CREATE /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MODIFY /home/remnux/.wine/reg14770000.tmp
10:03:25 MOVED_FROM /home/remnux/.wine/reg14770000.tmp
10:03:25 MOVED_TO /home/remnux/.wine/user.reg
```

Explanation: File System Activity

- Dropped uzico.exe into AppData/Roaming/Esbo/ - classic dropper behavior, spawning a secondary payload

- Created and executed tmp_65a9ca40.bat in AppData/Local/Temp/, then self-deleted it, consistent with the DeleteFileW behavior observed in static analysis
- Repeatedly modified reg14770000.tmp in the Wine root, likely registry manipulation or persistence mechanism writing in bulk

tcpdump:

```
remnux@remnux:~$ sudo tcpdump -i any -n -w /tmp/dontbother_capture.pcap 'not port 22'
tcpdump: data link type LINUX_SLL2
tcpdump: /tmp/dontbother_capture.pcap: Permission denied
remnux@remnux:~$ sudo tcpdump -i any -n -w /tmp/dontbother_capture.pcap 'not port 22'
tcpdump: data link type LINUX_SLL2
tcpdump: /tmp/dontbother_capture.pcap: Permission denied
remnux@remnux:~$ sudo tcpdump -i any -n -w dontbother_capture.pcap 'not port 22'
tcpdump: data link type LINUX_SLL2
tcpdump: listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
^C325 packets captured
337 packets received by filter
0 packets dropped by kernel
remnux@remnux:~$
```

Explanation:

- DNS query for 6pi3jrqjbssfh6gu.onion.pw observed on both loopback and outbound interface
- The .onion.pw domain is a Tor2Web proxy, meaning the malware is attempting C2 communication over Tor without requiring a Tor client on the victim machine
- Consistent with the tunneling behavioral tag flagged on VirusTotal, confirms active C2 beaconing behavior

Step19: Check the capture for onion:

19)Command: tcpdump -r dontbother_capture.pcap | GREP -iE onion

```
remnux@remnux:~$ tcpdump -r dontbother_capture.pcap | grep -iE onion
reading from file dontbother_capture.pcap, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144
Warning: interface names might be incorrect
10:03:07.111346 lo    In  IP localhost.32900 > _localdnsstub.domain: 29908+ [1au] A? 6pi3jrqjbssfh6gu.onion.pw. (54)
10:03:07.111810 enp0s3 Out IP remnux.56371 > 10.0.2.3.domain: 43559+ A? 6pi3jrqjbssfh6gu.onion.pw. (43)
remnux@remnux:~$
```

Step20: Inspect the onion domain:

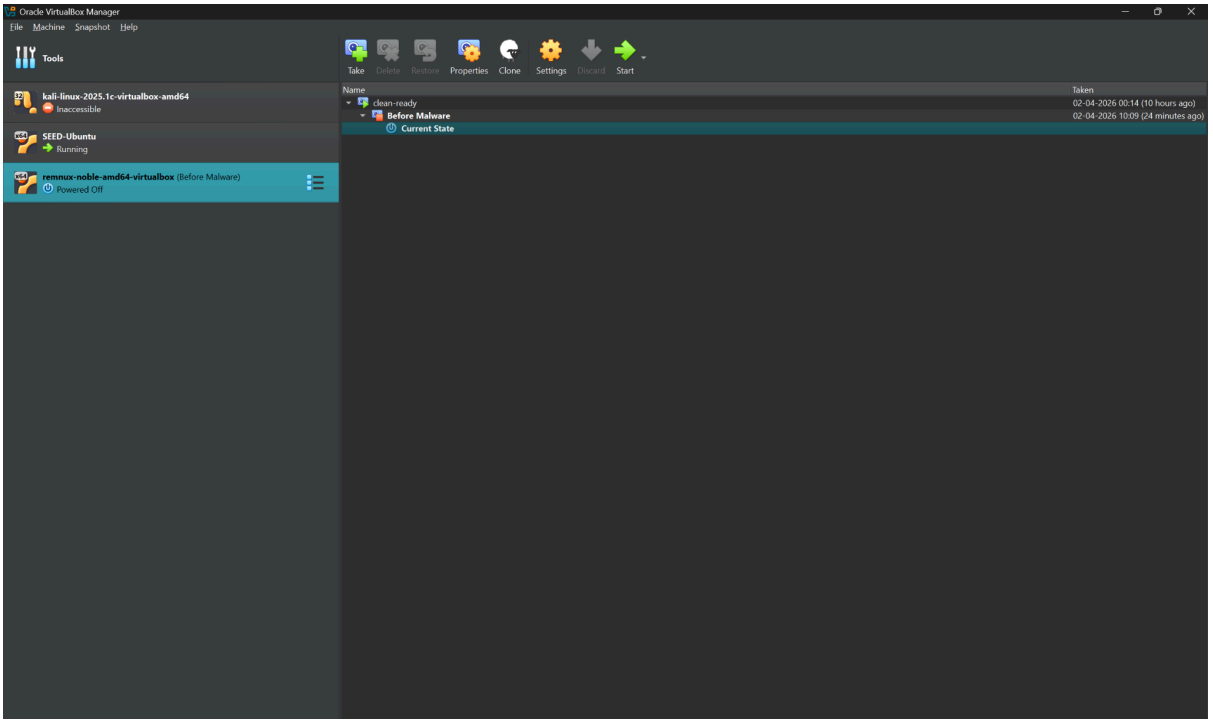
20) (Blocked by PES Network)

Explanation:

DNS query for 6pi3jrqjbssfh6gu.onion.pw issued at 10:03:07 on both loopback (localhost:32900) and outbound interface (remnux:56371 to 10.0.2.3)

- .onion.pw is a Tor2Web gateway, allowing Tor hidden service access without a Tor client, confirming C2 over Tor
- Dual interface query (loopback + external) suggests the DNS stub resolver forwarded the query outbound
- Domain structure (6pi3jrqjbssfh6gu) is a 16-character onion address, consistent with a v2 Tor hidden service used as a C2 endpoint

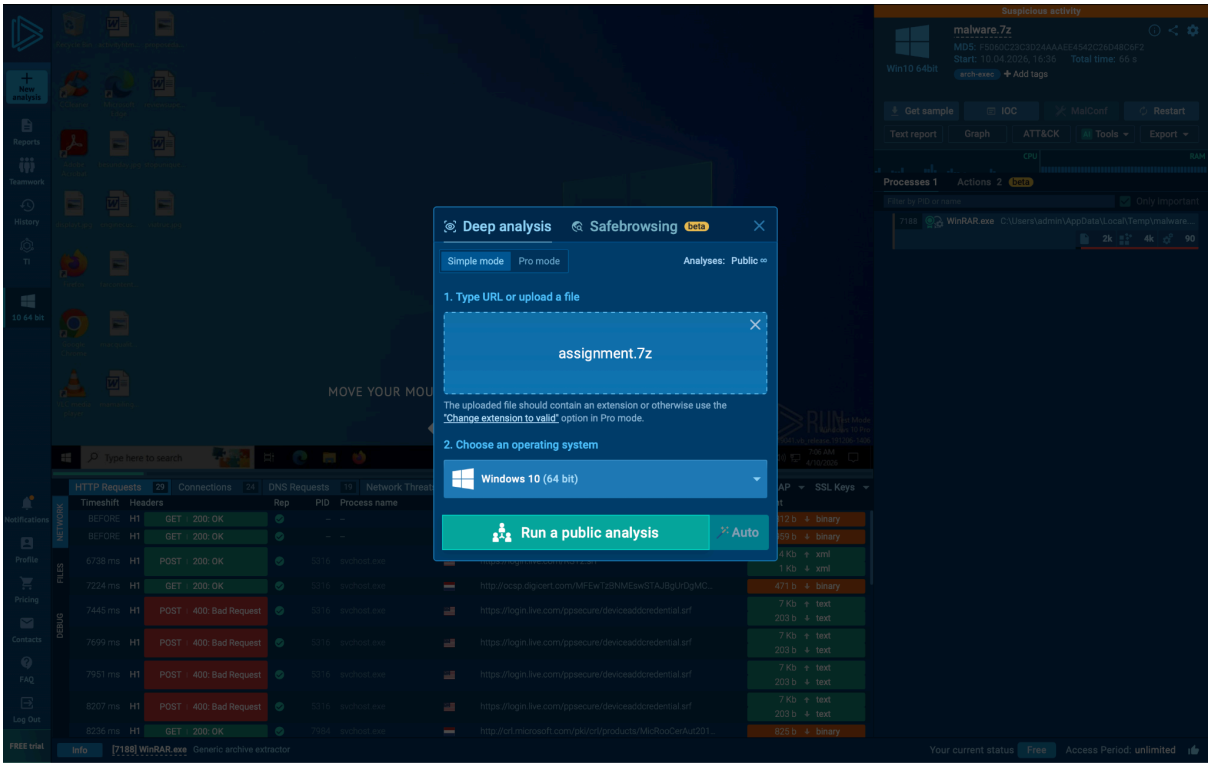
Step21: Restore snapshot:

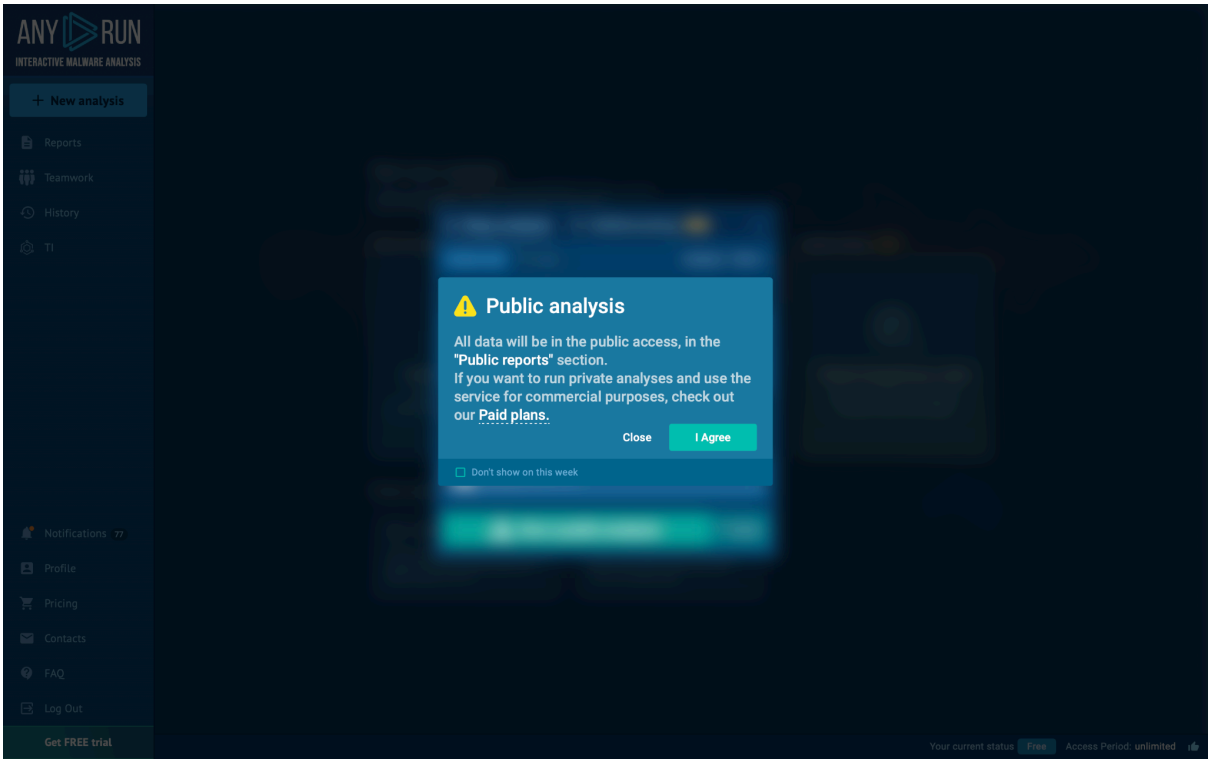


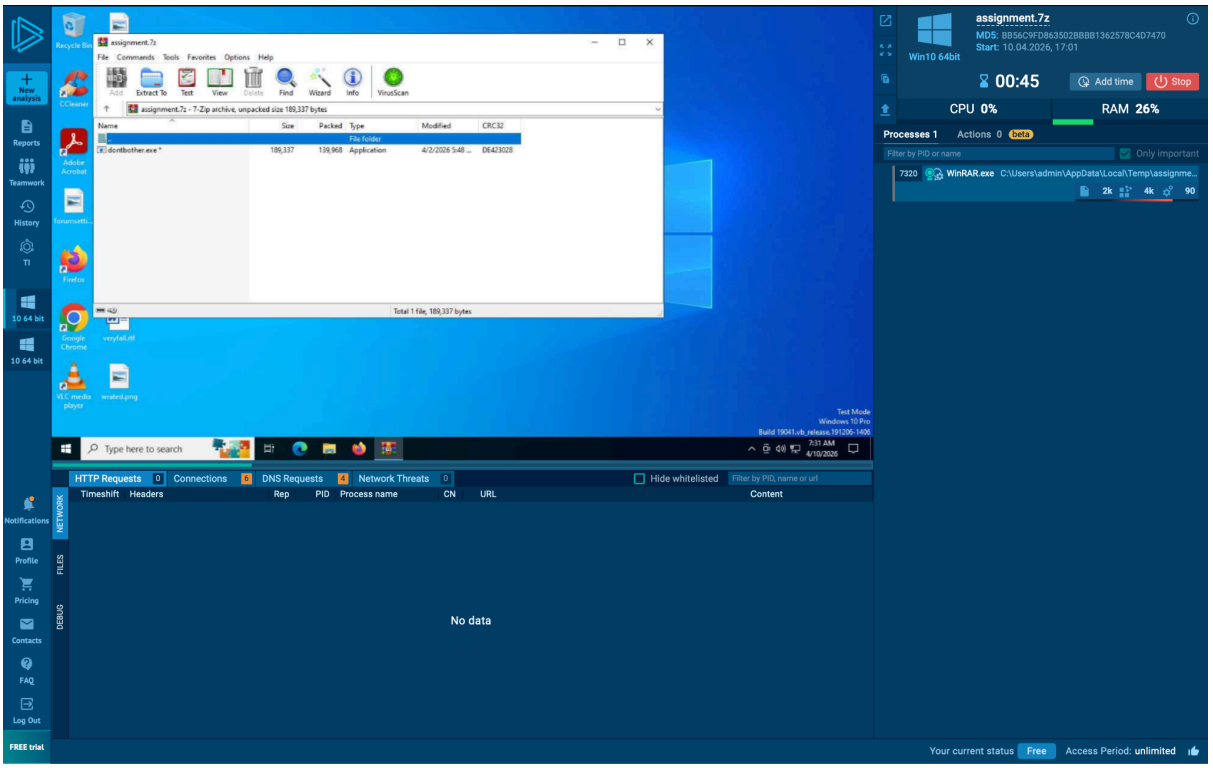
Step22: Do it online on anyrun:

Step23: Open the console:

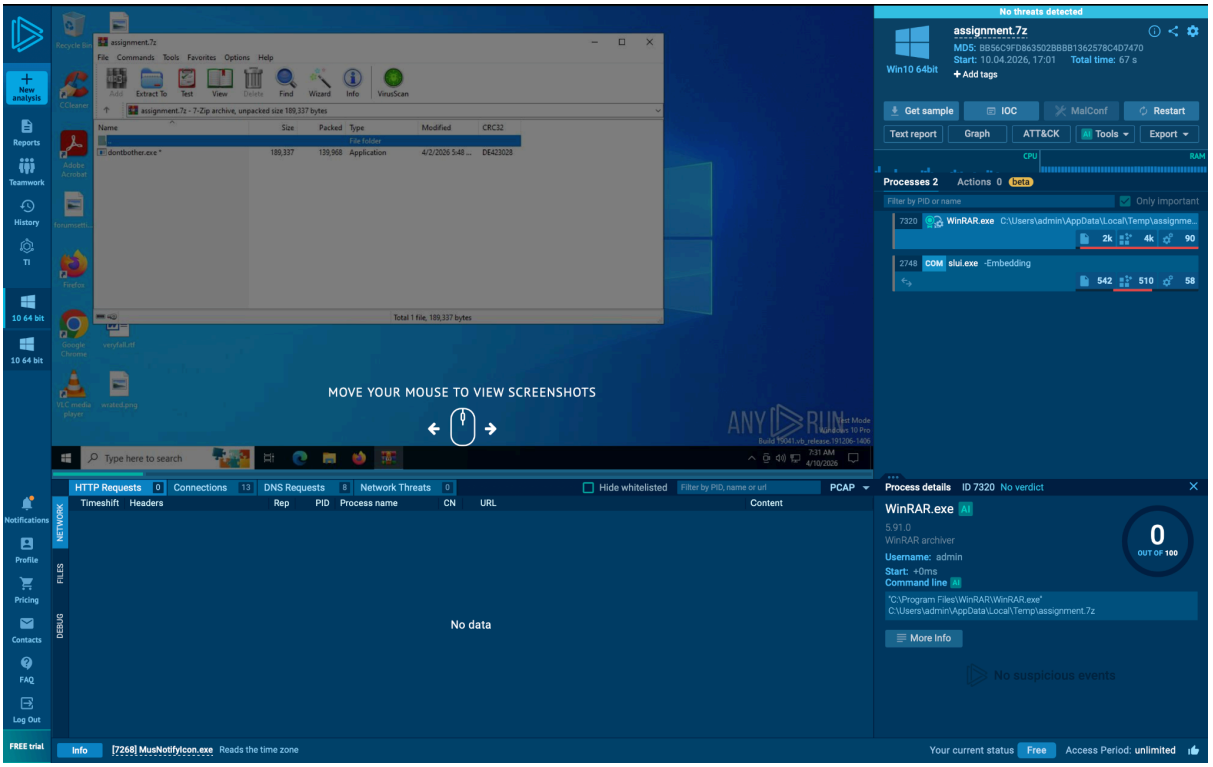
Step24: Upload the malware file to run analysis:

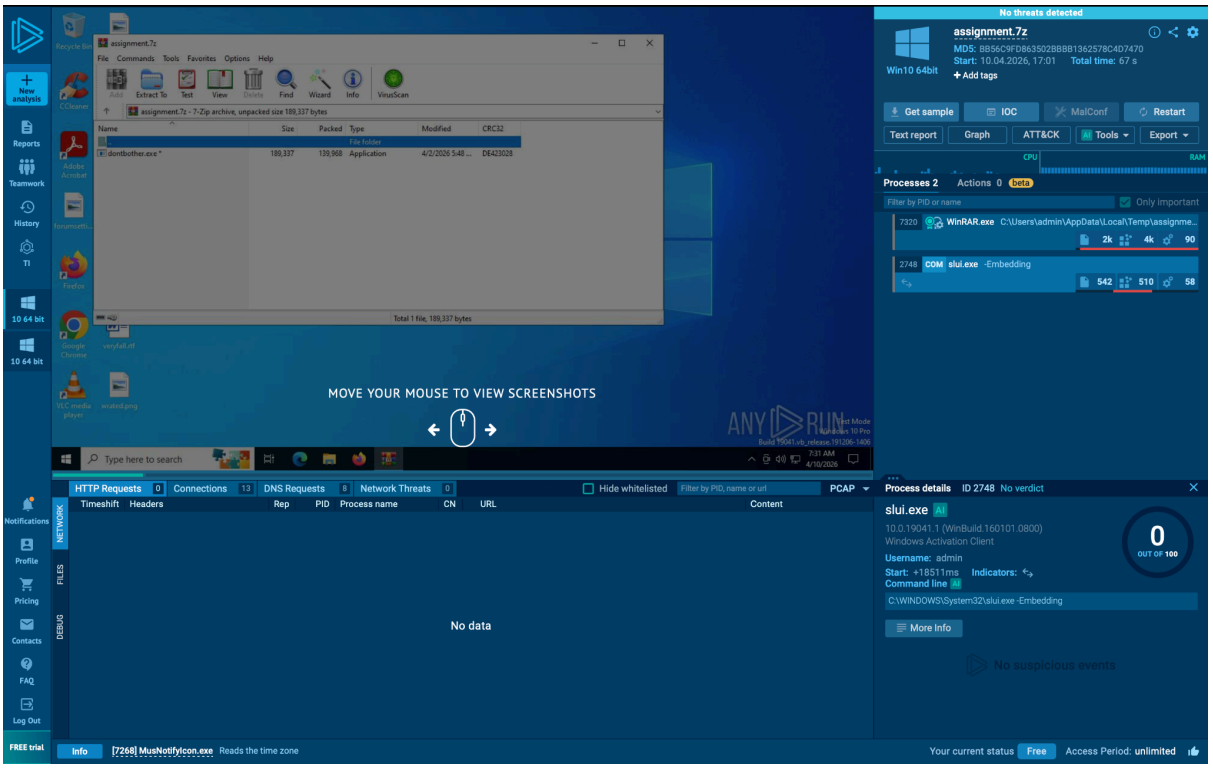






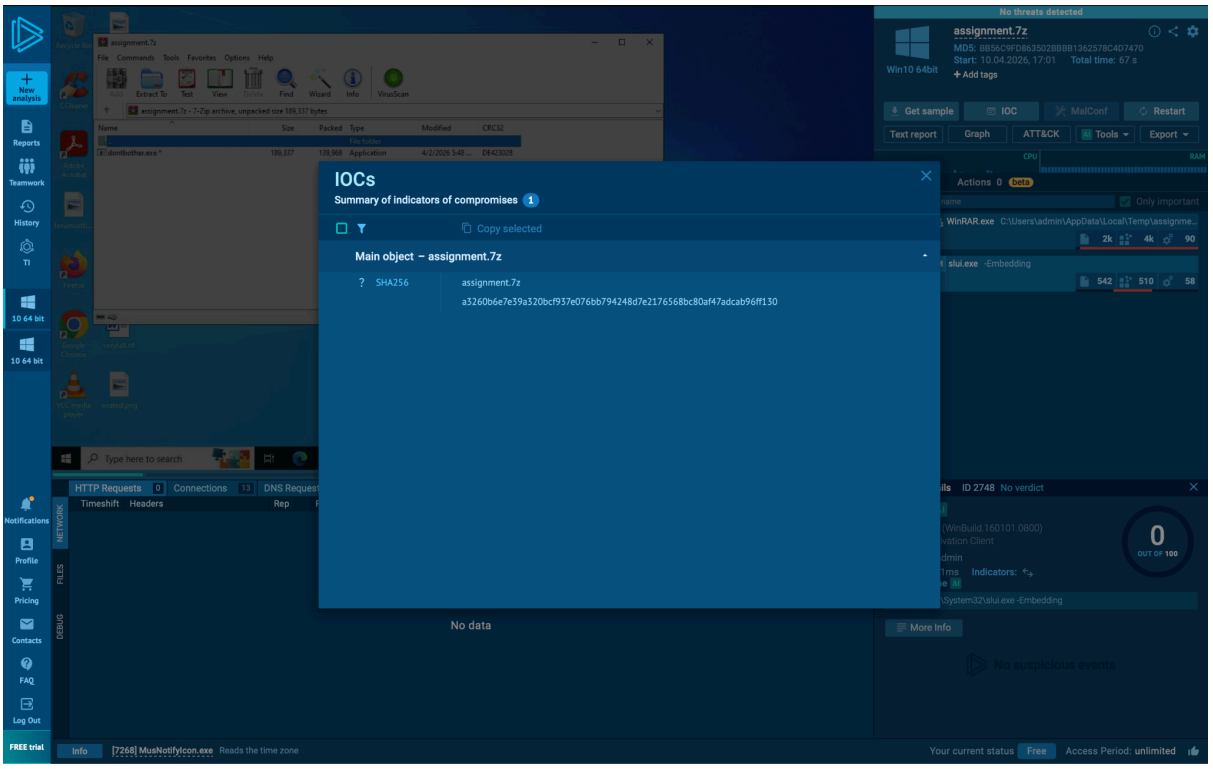
Step25: Observe the processes:





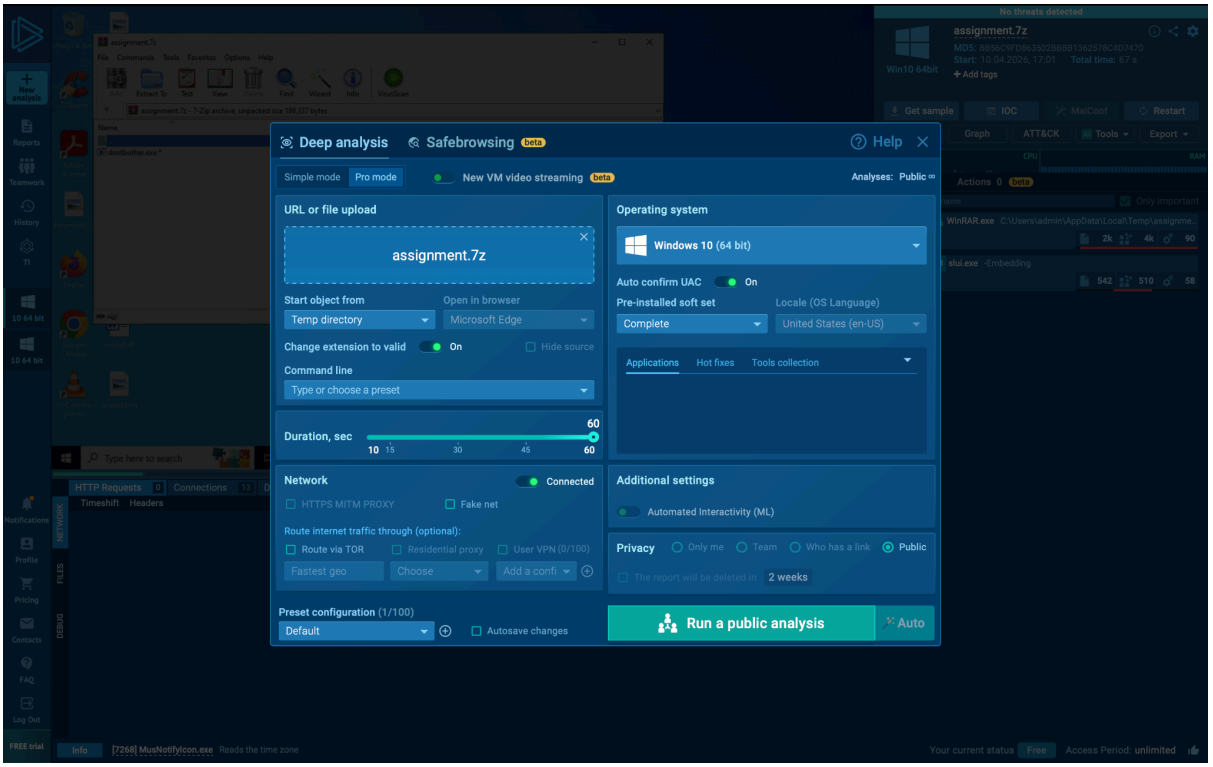
Explanation:
Anyrun allows us to inspect processes that ran for the uploaded file and as we can see they are two files - WinRAR.exe and stul.exe, it was not able to run dontbother.exe

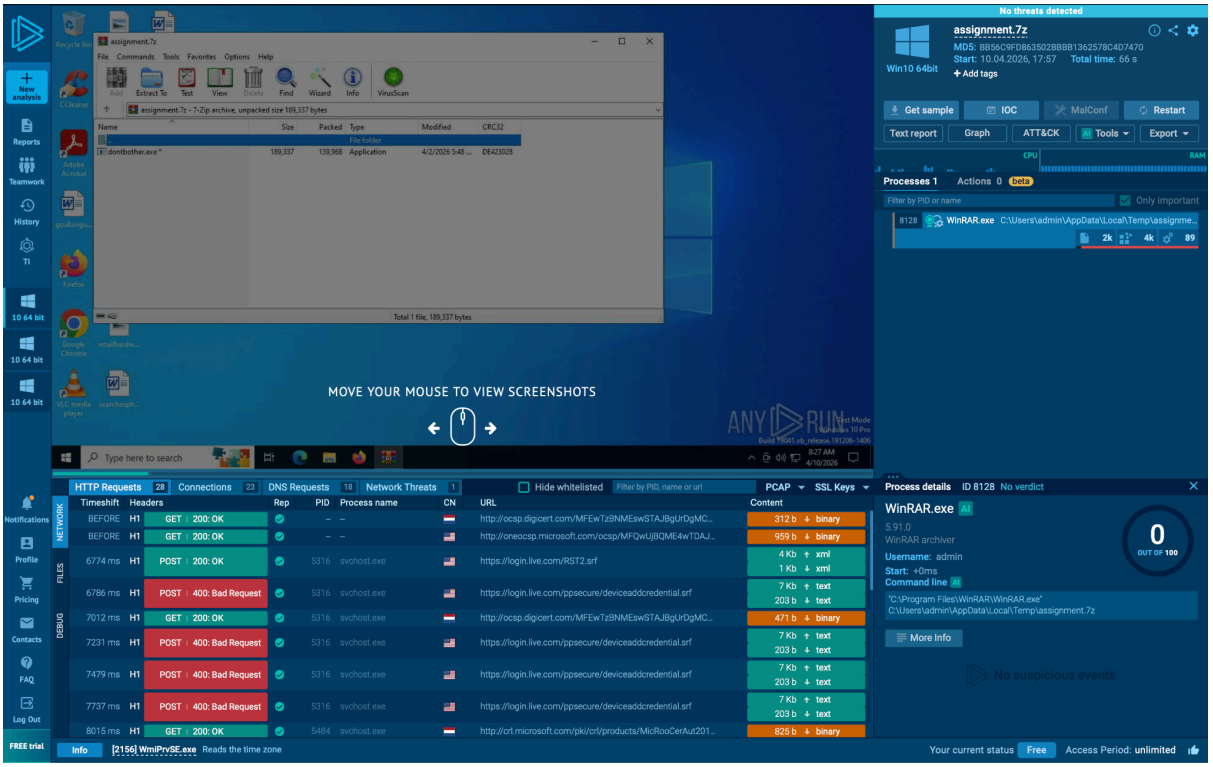
Step26: Observe the IOCs:



Explanation:
We observe no IOCs here as it didnt detect the dontbother.exe file

Step27: Rerun to try if it catches the malware:





Explanation:
As we observe from the screenshots, anyrun is unable to catch the malware dontbother.exe

Step28: Generate the analysis report:

ANY RUN
INTERACTIVE MALWARE ANALYSIS

General Info

File name:	assignment.7z
Full analysis:	https://app.any.run/tasks/4578024f-b40f-464a-ac3c-fcc22480a6dd
Verdict:	No threats detected
Analysis date:	April 10, 2026 at 17:01:39
OS:	Windows 10 Professional (build: 19044, 64 bit)
MIME:	application/x-7z-compressed
File info:	7-zip archive data, version 0.4
MD5:	BB56C9FD863502BBBB1362578C4D7470
SHA1:	41CE37A4CC0032950809E558D7C4DB1974595F0D
SHA256:	A3260B6E7E39A320BCF937E076BB794248D7E2176568BC80AF47ADCAB96FF130
SSDEEP:	3072:QGIRZhVBzvOu+MuxW6XZdGYC/oNGmsIV+H6x560cATCpdSv94CwSQI2R:QGkVBT0ZxWYQYCgNGaVJx5cATmQNkII

Software environment set and analysis options

Launch configuration

Task duration:	60 seconds	Heavy Evasion option:	off	Network geolocation:	off
Additional time used:	none	MITM proxy:	off	Privacy:	Public submission
Fakenet option:	off	Route via Tor:	off	Autoconfirmation of UAC:	on
Network:	on				

Software preset

- Internet Explorer 11.3636.19041.0
- Adobe Acrobat (64-bit) (23.001.20093)
- Adobe Flash Player 32 NPAPI (32.0.0.465)
- Adobe Flash Player 32 PPAPI (32.0.0.465)
- CCleaner (6.20)
- Google Chrome (133.0.6943.127)
- Google Update Helper (1.3.36.51)
- Java 8 Update 271 (64-bit) (8.0.2710.9)
- Java Auto Updater (2.8.271.9)
- Java(TM) SE Development Kit 25.0.2 (64-bit) (25.0.2.0)
- Microsoft Edge (133.0.3065.92)
- Microsoft Office Professional 2019 - de-de (16.0.16026.20146)
- Microsoft Office Professional 2019 - en-us (16.0.16026.20146)
- Microsoft Office Professional 2019 - es-es (16.0.16026.20146)
- Microsoft Office Professional 2019 - it-it (16.0.16026.20146)
- Microsoft Office Professional 2019 - ja-jp (16.0.16026.20146)
- Microsoft Office Professional 2019 - ko-kr (16.0.16026.20146)
- Microsoft Office Professional 2019 - pt-br (16.0.16026.20146)
- Microsoft Office Professional 2019 - tr-tr (16.0.16026.20146)
- Microsoft Office Professionnel 2019 - fr-fr (16.0.16026.20146)
- Microsoft Office профессиональный 2019 - ru-ru (16.0.16026.20146)
- Microsoft OneNote - en-us (16.0.16026.20146)
- Microsoft Update Health Tools (3.74.0.0)
- Microsoft Visual C++ 2013 Redistributable (x64) - 12.0.30501 (12.0.30501.0)
- Microsoft Visual C++ 2013 x64 Additional Runtime - 12.0.21005 (12.0.21005)
- Microsoft Visual C++ 2013 x64 Minimum Runtime - 12.0.21005 (12.0.21005)
- Microsoft Visual C++ 2015-2022 Redistributable (x64) - 14.36.32532 (14.36.32532.0)
- Microsoft Visual C++ 2015-2022 Redistributable (x86) - 14.36.32532 (14.36.32532.0)
- Microsoft Visual C++ 2022 X64 Additional Runtime - 14.36.32532 (14.36.32532)
- Microsoft Visual C++ 2022 X64 Minimum Runtime - 14.36.32532 (14.36.32532)
- Microsoft Visual C++ 2022 X86 Additional Runtime - 14.36.32532 (14.36.32532)
- Microsoft Visual C++ 2022 X86 Minimum Runtime - 14.36.32532 (14.36.32532)
- Mozilla Firefox (x64 en-US) (136.0)
- Mozilla Maintenance Service (136.0)
- Notepad++ (64-bit x64) (7.9.1)
- Office 16 Click-to-Run Extensibility Component (16.0.15726.20202)
- Office 16 Click-to-Run Licensing Component (16.0.16026.20146)
- Office 16 Click-to-Run Localization Component (16.0.15726.20202)
- Office 16 Click-to-Run Localization Component (16.0.15928.20198)

Hotfixes

- Client LanguagePack Package
- DotNetRollup
- DotNetRollup 481
- FodMetadata Package
- Foundation Package
- Hello Face Package
- InternetExplorer Optional Package
- KB5003791
- KB5011048
- KB5015684
- KB5033052
- LanguageFeatures Basic en us Package
- LanguageFeatures Handwriting en us Package
- LanguageFeatures OCR en us Package
- LanguageFeatures Speech en us Package
- LanguageFeatures TextToSpeech en us Package
- MSPaint FoD Package
- MediaPlayer Package
- Microsoft OneCore ApplicationModel Sync Desktop FOD Package
- Microsoft OneCore DirectX Database FOD Package
- NetFx3 OnDemand Package
- Notepad FoD Package
- OpenSSH Client Package
- PowerShell ISE FOD Package
- Printing PMCPPC FoD Package
- Printing WFS FoD Package
- ProfessionalEdition
- QuickAssist Package
- RollupFix
- ServicingStack
- ServicingStack 3989
- StepsRecorder Package
- TabletPCMath Package
- UserExperience Desktop Package
- WordPad FoD Package

- PowerShell 7-x64 (7.3.5.0)
- Update for Windows 10 for x64-based Systems (KB4023057) (2.59.0.0)
- Update for Windows 10 for x64-based Systems (KB4023057) (2.63.0.0)
- Update for Windows 10 for x64-based Systems (KB4480730) (2.55.0.0)
- Update for Windows 10 for x64-based Systems (KB5001716) (8.93.0.0)
- VLC media player (3.0.11)
- WinRAR 5.91 (64-bit) (5.91.0)
- Windows PC Health Check (3.6.2204.08001)

Behavior activities

MALICIOUS	SUSPICIOUS	INFO
No malicious indicators.	No suspicious indicators.	No info indicators.

Malware configuration

No Malware configuration.

Static information

TRiD

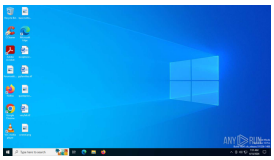
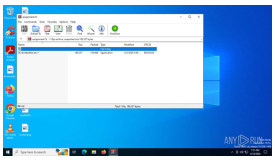
.7z | 7-Zip compressed archive (v0.4) (57.1)
.7z | 7-Zip compressed archive (gen) (42.8)

EXIF

ZIP

FileVersion:	7z v0.04
ModifyDate:	2026:04:02 09:48:46+00:00
ArchivedFileName:	dontbother.exe

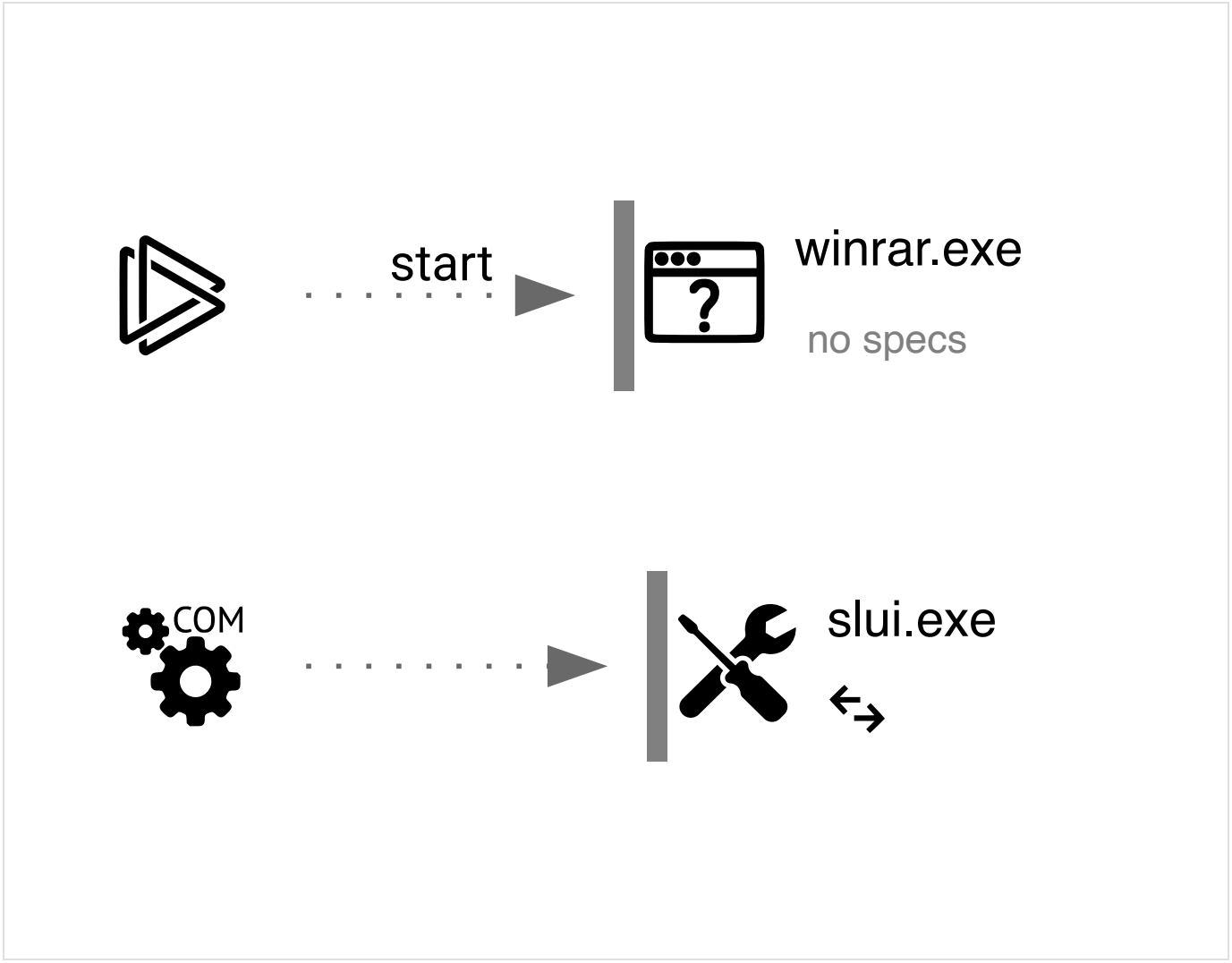
Video and screenshots

Processes

Total processes	Monitored processes	Malicious processes	Suspicious processes
133	2	0	0

Behavior graph



Specs description

- Program did not start
- Probably Tor was used
- Known threat
- Connects to the network
- Task contains several apps running
- File is detected by antivirus software
- The process has the malware config
- Low-level access to the HDD
- Behavior similar to spam
- RAM overrun
- CPU overrun
- Application downloaded the executable file
- Inspected object has suspicious PE structure
- Process was added to the startup
- Task has injected processes
- Network attacks were detected
- Process starts the services
- Actions similar to stealing personal data
- Behavior similar to exploiting the vulnerability
- Debug information is available
- Executable file was dropped
- Integrity level elevation
- System was rebooted
- Task has apps ended with an error
- Task contains an error or was rebooted

Process information

PID	CMD	Path	Indicators	Parent process
2748	C:\WINDOWS\System32\slui.exe -Embedding	C:\Windows\System32\slui.exe		svchost.exe
Information				

User:	admin	**Company:**	Microsoft Corporation
Integrity Level:	MEDIUM	**Description:**	Windows Activation Client
Exit code:	0	**Version:**	10.0.19041.1 (WinBuild.160101.0800)

7320 "C:\Program Files\WinRAR\WinRAR.exe" C:\Users\admin\AppData\Local\Temp\assignment.7z — C:\Program Files\WinRAR\WinRAR.exe — explorer.exe

Information

User:	admin	**Company:**	Alexander Roshal
Integrity Level:	MEDIUM	**Description:**	WinRAR archiver
Version:	5.91.0		

Registry activity

Total events	Read events	Write events	Delete events
2 009	2 000	9	0

Modification events

(PID) Process:	(7320) WinRAR.exe	**Key:**	HKEY_CURRENT_USER\SOFTWARE\WinRAR\ArcHistory
Operation:	write	**Name:**	3
Value:	C:\Users\admin\Desktop\chromium_ext.zip		
(PID) Process:	(7320) WinRAR.exe	**Key:**	HKEY_CURRENT_USER\SOFTWARE\WinRAR\ArcHistory
Operation:	write	**Name:**	2
Value:	C:\Users\admin\Desktop\omni_23_10_2024_.zip		
(PID) Process:	(7320) WinRAR.exe	**Key:**	HKEY_CURRENT_USER\SOFTWARE\WinRAR\ArcHistory
Operation:	write	**Name:**	1
Value:	C:\Users\admin\Downloads\chromium_build 1.zip		
(PID) Process:	(7320) WinRAR.exe	**Key:**	HKEY_CURRENT_USER\SOFTWARE\WinRAR\ArcHistory
Operation:	write	**Name:**	0
Value:	C:\Users\admin\AppData\Local\Temp\assignment.7z		
(PID) Process:	(7320) WinRAR.exe	**Key:**	HKEY_CURRENT_USER\SOFTWARE\WinRAR\FileList\FileColumnWidths
Operation:	write	**Name:**	name
Value:	120		
(PID) Process:	(7320) WinRAR.exe	**Key:**	HKEY_CURRENT_USER\SOFTWARE\WinRAR\FileList\FileColumnWidths
Operation:	write	**Name:**	size
Value:	80		
(PID) Process:	(7320) WinRAR.exe	**Key:**	HKEY_CURRENT_USER\SOFTWARE\WinRAR\FileList\FileColumnWidths
Operation:	write	**Name:**	type
Value:	120		
(PID) Process:	(7320) WinRAR.exe	**Key:**	HKEY_CURRENT_USER\SOFTWARE\WinRAR\FileList\FileColumnWidths
Operation:	write	**Name:**	mtime
Value:	100		
(PID) Process:	(2748) slui.exe	**Key:**	HKEY_CLASSES_ROOT\Local Settings\MuiCache\3d\52C64B7E
Operation:	write	**Name:**	@%SystemRoot%\System32\sppcomapi.dll,-3200
Value:	Software Licensing		

Files activity

Executable files	Suspicious files	Text files	Unknown types
0	0	0	0

Dropped files

No data

Network activity

HTTP(S) requests	TCP/UDP connections	DNS requests	Threats
0	13	8	0

HTTP requests

No HTTP requests

Connections

PID	Process	IP	Domain	ASN	CN	Reputation
4	System	192.168.100.255:137	—	Not routed	—	whitelisted
7452	svchost.exe	4.231.128.59:443	settings-win.data.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	whitelisted
5276	MoUsoCoreWorker.exe	4.231.128.59:443	settings-win.data.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	whitelisted
5412	slui.exe	48.192.1.65:443	activation-v2.sls.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	whitelisted
4	System	192.168.100.255:138	—	Not routed	—	whitelisted
7452	svchost.exe	51.124.78.146:443	settings-win.data.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	whitelisted
5276	MoUsoCoreWorker.exe	51.124.78.146:443	settings-win.data.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	whitelisted
2748	slui.exe	48.192.1.65:443	activation-v2.sls.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	whitelisted
5208	svchost.exe	51.124.78.146:443	settings-win.data.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK	US	whitelisted

DNS requests

Domain	IP	Reputation
settings-win.data.microsoft.com	4.231.128.59 51.124.78.146	whitelisted
activation-v2.sls.microsoft.com	48.192.1.65	whitelisted
google.com	142.251.13.138 142.251.13.102 142.251.13.100 142.251.13.139 142.251.13.113 142.251.13.101	whitelisted
self.events.data.microsoft.com	20.50.201.203	whitelisted

Threats

No threats detected

Debug output strings

No debug info

ANY.RUN INTERACTIVE MALWARE ANALYSIS
Interactive malware hunting service ANY.RUN
© 2017-2026 ANY.RUN ALL RIGHTS RESERVED

Step29: Generate a claude.ai analysis report: (static analysis)

MALWARE ANALYSIS REPORT

dontbother.exe

Prepared by: Pandemonium Research

Date: April 10, 2026

Classification: Confidential

1. Executive Summary

The submitted sample dontbother.exe is a 32-bit Windows PE executable compiled with MSVC 2013 in February 2017. Static and behavioral analysis reveals a tool with the hallmarks of a process-aware utility or low-sophistication implant: it enumerates running processes, hashes system identity information using the Windows CryptoAPI, reads and potentially executes files from disk, performs dynamic API resolution to evade static import analysis, and employs four distinct anti-debugging mechanisms.

No network indicators (C2 addresses, URLs, DNS names) were found in plaintext. Obfuscated string data is present and did not yield cleartext via simple single-byte XOR decoding. A large embedded binary blob in the .data section may represent an encrypted payload or configuration structure. The sample does not exhibit packer signatures but has borderline-high overall entropy (7.07), consistent with the presence of this blob.

Overall threat assessment: MEDIUM-HIGH. The combination of process enumeration with kill capability, cryptographic identity fingerprinting, self-deletion capability, and anti-debugging strongly suggests offensive intent, though the absence of network I/O imports limits its standalone persistence capability. It may be a component of a larger toolchain rather than a standalone implant.

2. File Metadata

Property	Value
Filename	dontbother.exe
File Size	189,337 bytes (185 KB)
MD5	c9c341eaf04c89933ed28cbc2739d325
SHA-256	1a0a2fd546e3c05e15b2db3b531cb8e8755641f5f1c17910ce2fb7bbce2a05b7
File Type	PE32 executable (GUI) Intel 80386, Windows
Architecture	x86 (32-bit)
Subsystem	Windows GUI (no console window)

Compile Timestamp	2017-02-23 19:28:24 UTC
Linker	MSVC 12.0 (Visual Studio 2013)
Target OS	Windows XP SP2+ (OS version 5.1)
Entry Point RVA	`0x13B9`
Image Base	`0x00400000`
Overall File Entropy	7.072 (borderline high)

3. Static Analysis

3.1 PE Header and Security Mitigations

Mitigation	Status	Notes
NX / DEP	Enabled	DllCharacteristics bit 0x0100 set
ASLR	Disabled	DllCharacteristics bit 0x0040 NOT set — fixed load address 0x400000
SEH	Enabled	Safe exception handling active
CFG	Not present	Control Flow Guard not used (VS2013)
Code Signing	None	No Authenticode signature

The absence of ASLR is notable. It means the binary loads at its preferred base 0x400000 on every run, making exploitation trivially reproducible and making static address references in the disassembly directly usable. This may be an oversight, or it may be intentional (e.g., the binary expects hardcoded addresses to work reliably when self-referencing its payload).

3.2 Sections

Section	Virtual Addr	Virtual Size	Raw Size	Entropy	Flags
.text	`0x1000`	`0x46D1`	`0x4800`	6.21	Execute / Read
.rdata	`0x6000`	`0x175C`	`0x1800`	6.08	Read
.data	`0x8000`	`0x12FF0`	`0x12800`	4.06	Read / Write
.reloc	`0x1B000`	`0x300`	`0x400`	5.26	Discardable / Read

Section entropies are within normal range for compiled code. The .data section (entropy 4.06) contains a very large hex-encoded binary blob embedded as a string literal. This blob runs several kilobytes and appears to be either an encrypted/encoded payload, a large configuration block, or embedded shellcode. Its exact purpose could not be determined without runtime tracing.

3.3 Import Analysis

The binary imports from three DLLs. Imports are grouped by behavioral cluster below.

Anti-debugging and exception handling:

Import	DLL	Call Site(s)	Purpose
IsDebuggerPresent	KERNEL32.dll	0x4035B7	Basic debugger detection
GetThreadContext	KERNEL32.dll	0x402216	Hardware breakpoint detection (DR registers)
AddVectoredExceptionHandler	KERNEL32.dll	0x4025A6	Register exception-based anti-debug handler
RemoveVectoredExceptionHandler	KERNEL32.dll	0x4025B4	Clean up handler post-check
SetUnhandledExceptionFilter	KERNEL32.dll	0x4022F7, 0x40262B, 0x403864	Override default exception handling
UnhandledExceptionFilter	KERNEL32.dll	0x40386D	Custom unhandled exception callback

Process enumeration and control:

Import	DLL	Call Site(s)	Purpose
CreateToolhelp32Snapshot	KERNEL32.dll	0x4019AC, 0x401A1F, 0x40226A	Snapshot all running processes (3 separate passes)
Process32FirstW	KERNEL32.dll	0x4019C2, 0x401A35, 0x402284	Start process list iteration
Process32NextW	KERNEL32.dll	0x4019ED, 0x401A58, 0x4022AC	Continue process list iteration
OpenProcess	KERNEL32.dll	0x4022CF	Open handle to target process
TerminateProcess	KERNEL32.dll	0x403857	Kill process by handle
FindWindowW	USER32.dll	—	Locate window by class/title name

Cryptographic operations (ADVAPI32):

Import	Call Site	Purpose
CryptAcquireContextW	0x40102D	Initialize crypto provider (RSA/AES detected in wide strings)
CryptCreateHash	0x401047	Create hash object (algorithm determined at runtime)
CryptHashData	0x40105B	Feed data into hash
CryptGetHashParam	0x401072	Retrieve computed hash value
CryptDestroyHash	0x401088	Clean up hash object
CryptReleaseContext	0x401093	Release crypto provider
GetUserNameW	0x401F51	Retrieve current username (likely hashed for fingerprinting)

File I/O and execution:

Import	Call Site(s)	Purpose
CreateFileW	`0x40119F`	Open file for reading
GetFileSize	`0x4011AF`	Determine file size
VirtualAlloc	`0x4011D1`	Allocate memory for file contents
ReadFile	`0x4011E9`	Read file into allocated buffer
VirtualFree	`0x401203, 0x40149B`	Free allocated buffers
CreateProcessW	—	Spawn a new process (likely executes read payload)
DeleteFileW	—	Self-deletion or cleanup of dropped files
GetModuleHandleW/A	6 sites	Resolve base addresses of loaded DLLs dynamically
GetProcAddress	6 sites	Resolve function addresses at runtime (IAT evasion)
LoadLibraryW	`0x4023F5`	Load additional DLL at runtime

3.4 String Analysis

No plaintext network indicators (URLs, IP addresses, hostnames, registry paths) were found. The binary contains multiple short, garbled ASCII sequences that resist single-byte XOR decoding, suggesting either multi-byte keying, a custom substitution cipher, or runtime-generated strings. The wide (UTF-16) string table contains exclusively Windows CryptoAPI provider names, indicating the binary explicitly selects the Microsoft Enhanced RSA and AES Cryptographic Provider at runtime. A large hex-encoded data blob (several thousand bytes) is embedded in the .data section as a null-terminated ASCII string.

4. Behavioral Analysis (Static Simulation)

4.1 Entry Point and Initialization

Disassembly of the entry point at 0x4013B9 reveals a standard MSVC CRT startup sequence: stack frame allocation, heap initialization via GetProcessHeap, and GetModuleHandleW to resolve the base address of the loaded image. A conditional branch at 0x4013F1 routes execution toward 0x401707 (which calls ExitProcess) on failure. The second major initialization call at 0x4013F7 is an internal function at 0x4023CF — likely the main logic dispatcher. This is consistent with a WinMain()-style entry where pre-checks gate execution.

4.2 Anti-Debugging Sequence

The binary employs four layered anti-debugging mechanisms:

- IsDebuggerPresent (0x4035B7): Basic PEB-based check. Simple to bypass via PEB patching but present as a first gate.
- GetThreadContext (0x402216): Retrieves the thread context of the current thread to inspect debug registers DR0-DR3. Non-zero DR values indicate hardware breakpoints set by a debugger.
- AddVectoredExceptionHandler / SetUnhandledExceptionFilter: Registers custom exception handlers. Techniques using int3 or privileged instruction traps as anti-debug checks are common in this pattern — if a debugger intercepts the exception, the handler never fires and execution takes a different path.

- Three separate calls to SetUnhandledExceptionFilter suggest the binary re-asserts exception control at multiple points in execution flow, a sign of deliberate anti-debug layering rather than incidental CRT behavior.

4.3 Process Enumeration

Three separate process enumeration loops are present, each using the CreateToolhelp32Snapshot / Process32First / Process32Next pattern. The distinct call sites (0x4019xx, 0x401Axx, 0x4022xx) suggest different purposes per pass — for example, one pass to locate a target process by name, a second to verify termination, and a third to enumerate for security tools. OpenProcess followed by TerminateProcess provides kill capability. FindWindowW may supplement process lookup by searching for a specific window title.

4.4 Cryptographic Fingerprinting

The sequenced call pattern CryptAcquireContextW -> CryptCreateHash -> CryptHashData -> CryptGetHashParam -> CryptDestroyHash -> CryptReleaseContext at 0x40102D-0x401093 constitutes a complete hash computation pipeline. GetUserNameW at 0x401F51 retrieves the current logged-in username, which is the most likely input to this hash. Wide string analysis confirms the Microsoft Enhanced RSA and AES Cryptographic Provider is explicitly requested. The resulting hash value is most plausibly used as a machine-specific key for decrypting the embedded payload blob, or as a system fingerprint included in an output artifact.

4.5 File Read and Execute Pattern

The sequence CreateFileW -> GetFileSize -> VirtualAlloc -> ReadFile at 0x40119F-0x4011E9 reads a file from disk into a heap-allocated buffer. This is followed by VirtualFree for cleanup. CreateProcessW is imported and GetModuleHandleW/GetProcAddress are used extensively for dynamic resolution, meaning additional capabilities may be invoked via APIs not visible in the static import table. DeleteFileW provides a self-deletion path, consistent with a dropper that removes itself after delivering a payload.

4.6 Dynamic Import Resolution

GetModuleHandleA/W and GetProcAddress are called at six distinct sites each, with LoadLibraryW at 0x4023F5. This is significantly more dynamic resolution than typical benign software. Combined with the direct RtlGetVersion call through ntdll.dll, the binary resolves a substantial portion of its actual functionality at runtime, rendering the static import table incomplete as a behavioral indicator.

5. Findings Summary

ID	Severity	Finding	Evidence
F-01	HIGH	Anti-debugging (4 mechanisms)	IsDebuggerPresent, GetThreadContext, VEH manipulation, SEH override
F-02	HIGH	Process enumeration with kill capability	3x toolhelp loops, OpenProcess + TerminateProcess
F-03	HIGH	Dynamic import resolution (IAT evasion)	6x GetProcAddress / GetModuleHandle call sites + LoadLibraryW
F-04	HIGH	Large encrypted/encoded embedded payload	Multi-KB hex blob in .data section; unresolved encoding

F-05	MEDIUM	Cryptographic identity fingerprinting	CryptoAPI + GetUserNameW pipeline; key derivation likely
F-06	MEDIUM	File read-execute pattern (dropper behavior)	CreateFile/ReadFile/VirtualAlloc/CreateProcess chain
F-07	MEDIUM	Self-deletion capability	DeleteFileW imported and call sites present
F-08	MEDIUM	Obfuscated string table	Multiple garbled ASCII strings; cipher not identified
F-09	LOW	ASLR disabled	Fixed load address 0x400000; increases exploit reproducibility
F-10	LOW	No version info or resources	No ProductName, CompanyName, FileDescription — typical of malware
F-11	INFO	Compiled 2017 with VS2013	Compile timestamp 2017-02-23; older toolchain
F-12	INFO	No network I/O imports	No Winsock, WinInet, WinHTTP — limits standalone C2 capability

6. Threat Assessment

Overall Rating: MEDIUM-HIGH

The sample presents a coherent offensive capability profile: it detects analysis environments, enumerates and can kill processes, fingerprints the host via cryptographic hashing, reads and potentially executes files, and implements multiple layers of string and API obfuscation. This is not consistent with benign software behavior.

The absence of network imports is the primary factor preventing a HIGH rating. Without Winsock, WinInet, or WinHTTP, the binary cannot independently establish outbound connections, suggesting it either: (a) operates as a component in a multi-stage toolchain where another module handles C2 communication, (b) communicates via a loaded DLL resolved through the dynamic import mechanism, or (c) is a utility executed by a separate loader that passes it instructions via file or process injection.

The embedded payload blob is the most significant unresolved artifact. Its decryption is gated on the cryptographic hash derived from host-specific data (username), making static decryption impossible without knowing the target environment. This is a deliberate keying strategy that prevents off-system analysis.

7. Recommendations

- Dynamic analysis required: Execute in an instrumented sandbox (e.g., Cuckoo, ANY.RUN) on a VM with the same username as the suspected target to trigger payload decryption.
- Monitor for process kill behavior: In a live environment, watch for unexpected process terminations following execution of this binary.
- Patch point for payload extraction: Set a breakpoint at the CryptGetHashParam return (0x401072) to capture the derived key, then dump the decoded payload from memory.
- Check for companion files: The CreateFileW call at 0x40119F likely opens a hardcoded or command-line-specified path. Capture GetCommandLineW at 0x40141A to identify the expected input file.
- Isolate immediately: Do not run on a production or networked machine. The process-kill capability and self-deletion mean forensic recovery post-execution may be limited.

- Submit to threat intel platforms: SHA-256 1a0a2fd546e3c05e15b2db3b531cb8e8755641f5f1c17910ce2fb7bbce2a05b7 should be checked against VirusTotal, MalwareBazaar, and internal TI feeds.

Appendix A: Disassembly — Entry Point (first 30 instructions)

Address	Mnemonic	Operands	Notes
0x4013B9	sub	esp, 0x62C	Allocate 1580-byte stack frame
0x4013BF	call	[0x406058]	GetProcessHeap
0x4013C5	and	[0x41AC00], 0	Zero init global
0x4013CC	lea	edx, [esp+0x30]	
0x4013D0	and	[0x41AC04], 0	Zero init global
0x4013D7	push	0x2B	
0x4013D9	pop	ecx	ecx = 43
0x4013DA	mov	[0x41AC38], eax	Store heap handle
0x4013DF	call	0x40299C	Internal init function
0x4013E4	lea	eax, [esp+0x30]	
0x4013E9	call	[0x406050]	GetModuleHandleW (NULL = self)
0x4013EF	test	eax, eax	Check if loaded
0x4013F1	jne	0x401707	Jump to ExitProcess if fail
0x4013F7	call	0x4023CF	Main logic dispatcher
0x4013FC	and	[esp], 0	
0x401400	push	0x24	
0x401402	pop	eax	eax = 36
0x401403	dec	eax	Loop counter
0x401404	mov	byte [esp+eax+8], 0	Zero-fill buffer
0x401409	jne	0x401403	Loop until done
0x40140B	push	0x104	MAX_PATH = 260
0x401410	lea	eax, [esp+0x428]	Path buffer address
0x401417	push	eax	
0x401418	push	0	hModule = NULL (self)
0x40141A	call	[0x40606C]	GetModuleFileNameW — get own path
0x401420	push	ecx	

0x401421	lea	edx, [esp+0x34]	
0x401425	lea	ecx, [esp+0x428]	
0x40142C	call	0x401188	Internal path processing
0x401431	test	eax, eax	

Appendix B: Hashes

Algorithm	Hash
MD5	c9c341eaf04c89933ed28cbc2739d325
SHA-256	1a0a2fd546e3c05e15b2db3b531cb8e8755641f5f1c17910ce2fb7bbce2a05b7